nature machine intelligence

Article

https://doi.org/10.1038/s42256-025-01150-3

Learning cell dynamics with neural differential equations

Received: 24 January 2025

Accepted: 30 October 2025

Published online: 18 December 2025

Check for updates

Michael E. Vinyard[1,2,3], Anders W. Rasmussen[2], Ruitong Li[2,4], Allon M. Klein[5], Gad Getz[2,3,6,7,9] & Luca Pinello[2,3,6,7,8,9]

Single-cell sequencing measurements facilitate the reconstruction of dynamic biology by capturing snapshot molecular profiles of individual cells. Cell fate decisions in development and disease are orchestrated through an intricate balance of deterministic and stochastic regulatory events. Drift-diffusion equations are effective in modelling single-cell dynamics from high-dimensional single-cell measurements. While existing solutions describe the deterministic dynamics associated with the drift term of these equations at the level of cell state, diffusion is modelled as a constant across cell states. To fully understand the dynamic regulatory logic in development and disease, models explicitly attuned to the balance between deterministic and stochastic biology are required. To address these limitations, we introduce scDiffEq, a generative framework for learning neural stochastic differential equations that approximate biology's deterministic and stochastic dynamics. Using lineage-traced single-cell data, we demonstrate that scDiffEq offers an improved reconstruction of cell trajectories and prediction of cell fate from multipotent progenitors during haematopoiesis. By imparting in silico perturbations to multipotent progenitor cells, we find that scDiffEq accurately recapitulates the dynamics of CRISPR-perturbed haematopoiesis. We generalize this approach beyond lineage-traced or multi-time-point datasets to model the dynamics of single-cell data from a single time point. Using scDiffEq, we simulate high-resolution developmental cell trajectories, which can model their drift and diffusion, enabling us to study their time-dependent gene-level dynamics.

Dynamical systems underpin fundamental processes in biology and disease, including differentiation and cancer. Gene expression serves as a molecular proxy to characterize cell states. Single-cell RNA sequencing (scRNA-seq) captures snapshots of both stable and transient cell states as subpopulations transition between them. While individual cells are destroyed on measurement, scRNA-seq profiles thousands of cells, enabling computational inference of relationships between a cell's observed state and its past or future states, offering insights into regulatory dynamics.

Computational tools for studying single-cell dynamics have grown increasingly sophisticated[1–6]. Trajectory inference methods order cells along pseudotemporal axes but rely on correlative associations, offering limited insight into the underlying mechanisms driving transitions. RNA velocity leverages assumptions about the kinetics of RNA transcription, splicing and degradation to infer future cell state from unspliced and spliced transcript abundances[7,8]. Methods, including Dynamo and CellRank (and recently, CellRank2), use RNA velocity to infer long-range cell trajectories and fates[9–11]. While RNA velocity-based

[1]Department of Chemistry and Chemical Biology, Harvard University, Cambridge, MA, USA. [2]Broad Institute of MIT and Harvard, Cambridge, MA, USA. [3]Department of Pathology, Massachusetts General Hospital, Boston, MA, USA. [4]Department of Biomedical Informatics, Harvard Medical School, Cambridge, MA, USA. [5]Department of Systems Biology, Harvard Medical School, Boston, MA, USA. [6]Krantz Family Center for Cancer Research, Massachusetts General Hospital, Boston, MA, USA. [7]Department of Pathology, Harvard Medical School, Boston, MA, USA. [8]Department of Biostatistics, Harvard TH Chan School of Public Health, Boston, MA, USA. [9]These authors jointly supervised this work:Gad Getz, Luca Pinello.
e-mail: gadgetz@broadinstitute.org; lpinello@mgh.harvard.edu

https://doi.org/10.1038/s42256-025-01150-3

methods model the mean drift of cell states, they do not incorporate cell-specific diffusion and are sensitive to data preprocessing and biophysical assumptions around transcriptional kinetics[12].

Drift-diffusion equations model cell dynamics from snapshot data (Fig. 1a), but analytical solutions are intractable in high dimensions (>1,000 genes). Early frameworks required strong assumptions. Population balance analysis (PBA) models steady-state cell dynamics via spectral graph theory and weighted random walks[13]. PRESCIENT, a generative recurrent neural network, learns a drift field from coarse time-resolved data using an optimal transport loss, regularized by the assumption that cells follow a gradient potential landscape (that is, model drift is the negative gradient of the potential), reflecting a Waddington landscape[14,15]. Dynamo infers a smoothed drift field from noisy velocity estimates[9]. Recent approaches based on neural ordinary differential equations (scTour, TIGON, TrajectoryNet, MIOFlow) model dynamics over time[16–19]. WaddingtonOT, PRESCIENT and TIGON further incorporate cell growth and death[6,15,17]. While PBA, PRESCIENT and Dynamo adopt drift-diffusion frameworks, they constrain diffusion to homogeneous, isotropic Gaussian noise, treating its magnitude as a dataset-level hyperparameter[9,13,15]. This constraint assumes stochastic dynamics are cell state-independent, limiting insights into how gene expression noise varies with cell state.

Stochasticity is essential for generating diverse cell types from a common progenitor and works alongside deterministic regulatory mechanisms during development[20,21]. Interpreting the interplay between stochastic and deterministic gene expression is vital to modelling complex cell decision-making. We aimed to clarify how cells exploit stochasticity to coordinate transcriptional changes over time and across gene expression space. Historically, differential equations have served as a foundational workhorse for biological modelling; however, they are typically restricted to low-dimensional data and require assumptions built on decades of empirical observations. Recent advances in deep learning, particularly neural differential equations, enable numerical approximation of complex dynamics[22–24]. Neural stochastic differential equations (neural SDEs) offer a direct parameterization of a drift-diffusion equation with deep neural networks[23] (Fig. 1b), providing a powerful framework for modelling single-cell dynamics. A physics-informed neural SDE framework, PI-SDE was independently proposed to model the dynamics of time-series scRNA-seq data[25]. PI-SDE was primarily demonstrated on data interpolation. In contrast, we demonstrate an approach to model cell dynamics that extends beyond interpolation to include fate prediction and functional analyses, such as in silico perturbation.

We build on existing models of cell dynamics, taking advantage of neural differential equations, to develop scDiffEq, a deep learning framework that learns neural SDEs from embeddings of transcriptional cell states to model and study their deterministic and stochastic dynamics (Fig. 1c). We benchmark scDiffEq using multi-time-point lineage-traced scRNA-seq data (Fig. 1d–f), noting improved accuracy in predicting dominant cell fates from multipotent progenitors compared with existing methods. scDiffEq also accurately interpolates distributions of held-out cell populations and identifies genes correlated with drift and diffusion along simulated trajectories. Focusing on the granulocyte–monocyte progenitor (GMP) transition to neutrophils and monocytes, we recover temporal dynamics of key transcription factors (TFs) and observe distinct diffusion patterns, which we propose are essential for maintaining non-equilibrium transcriptional dynamics in mouse haematopoiesis. This work highlights the importance of modelling diffusion for understanding cellular dynamics and offers a framework for its biological interpretation.

Results

Learning neural differential equations with scDiffEq

scDiffEq learns neural SDEs from principal component analysis (PCA)-reduced transcriptional profiles by simulating cell state trajectories from an initial population and minimizing the Sinkhorn divergence (regularized Wasserstein distance) between predicted and observed cell populations (Fig. 1c and Methods). To illustrate scDiffEq, we used a lineage-traced scRNA-seq dataset profiling mouse haematopoiesis with the lineage and RNA recovery (LARRY) barcoding system, measured at days 2, 4 and 6 posttransduction[26]. This dataset preserves state–state and state–fate relationships across time points, despite destructive measurement.

scDiffEq advances day 2 cells in discrete time steps, minimizing Sinkhorn divergence between predicted and observed populations at days 4 and 6 to fit the neural SDE (Methods). Simulated cells are annotated via nearest neighbours, with 50,000 trajectories capturing 85.5 ± 1.0% of the dataset manifold (Supplementary Fig. 1). Manifold-based outlier detection confirmed that 98.3 ± 1.1% of simulated trajectories contained biologically plausible states with minimal divergence at lowly sampled regions of the cell manifold (Supplementary Fig. 2).

Benchmarking models with lineage-traced scRNA-seq data

Benchmark datasets with standardized tasks are essential for validating predictive models. However, because single-cell assays destroy the measured cell, true past and future states remain unobservable. The LARRY dataset circumvents this limitation by transducing multipotent progenitor cells with heritable barcodes, enabling lineage tracing over time. By dividing cells into parallel wells, LARRY provides a natural train–test split (Fig. 2a–c and Supplementary Fig. 3a,b). As part of a growing class of lineage-traced datasets, LARRY offers a unique opportunity to approximate temporal dynamics including state–state and state–fate relationships[26–33].

We adapted benchmark tasks from previous studies to evaluate scDiffEq on fate prediction (task 1, Fig. 1d), interpolation of unobserved time points (task 2, Fig. 1e and Supplementary Fig. 3c) and progenitor perturbation prediction[9,15,17,26] (task 3, Fig. 1f).

In task 1, we benchmarked fate-prediction accuracy using lineage tracing as ground truth[15] (Fig. 2c and Methods). Linear regression achieved 61.9 ± 0.2% accuracy, reflecting that fate decisions are largely prefigured by static signatures (Supplementary Fig. 4), but cannot model underlying dynamics. In contrast, generative models must accomplish the additional task of modelling and reconstructing dynamic cellular processes over time, providing mechanistic biological insights.

Existing methods achieved 4.1–46.1% accuracy. Notably, RNA velocity-based approaches (Dynamo, CellRank) performed poorly. PRESCIENT achieves 42.5 ± 1.2% accuracy and this performance is boosted to 50.7 ± 1.2% (n = 5) when using the Kyoto Encyclopedia of Genes and Genomes (KEGG) gene expression signature to modulate expected cell growth in the optimal transport loss[6,15] (Fig. 2d).

scDiffEq outperformed all single-cell-specific methods with 52.5 ± 1.3% (n = 5, no KEGG) and 52.9 ± 4.9% (with KEGG), a ~2.2% improvement over the state of the art (Fig. 2d). These results indicate that scDiffEq is less dependent on growth weights than PRESCIENT. In addition, modelling non-uniform diffusion enables more accurate fate prediction outside dominant fates such as neutrophils and monocytes (Supplementary Fig. 5).

We examined how cell change ($d\mathbf{x}$) was distributed between drift and diffusion. We observed diffusion-dominated dynamics (drift/diffusion ratio 0.37 ± 0.14) versus PRESCIENT's extremely high ratio of 161.2 (Supplementary Fig. 6), with both approaches producing suboptimal entropy patterns compared with real lineage entropy (0.11). Systematic evaluation revealed optimal performance at drift/diffusion ratio 2.5 (accuracy 54.8 ± 2.6%), representing a ~6.1% improvement over the state of the art (Supplementary Figs. 7–8 and Methods).

Fate-prediction performance remained stable between ratios 1.0–5.0, peaking at 2.5 (accuracy 54.8 ± 2.6%), with sharp performance drops at 10.0 and above (Fig. 2d and Supplementary Fig. 8a,b). This

 https://doi.org/10.1038/s42256-025-01150-3

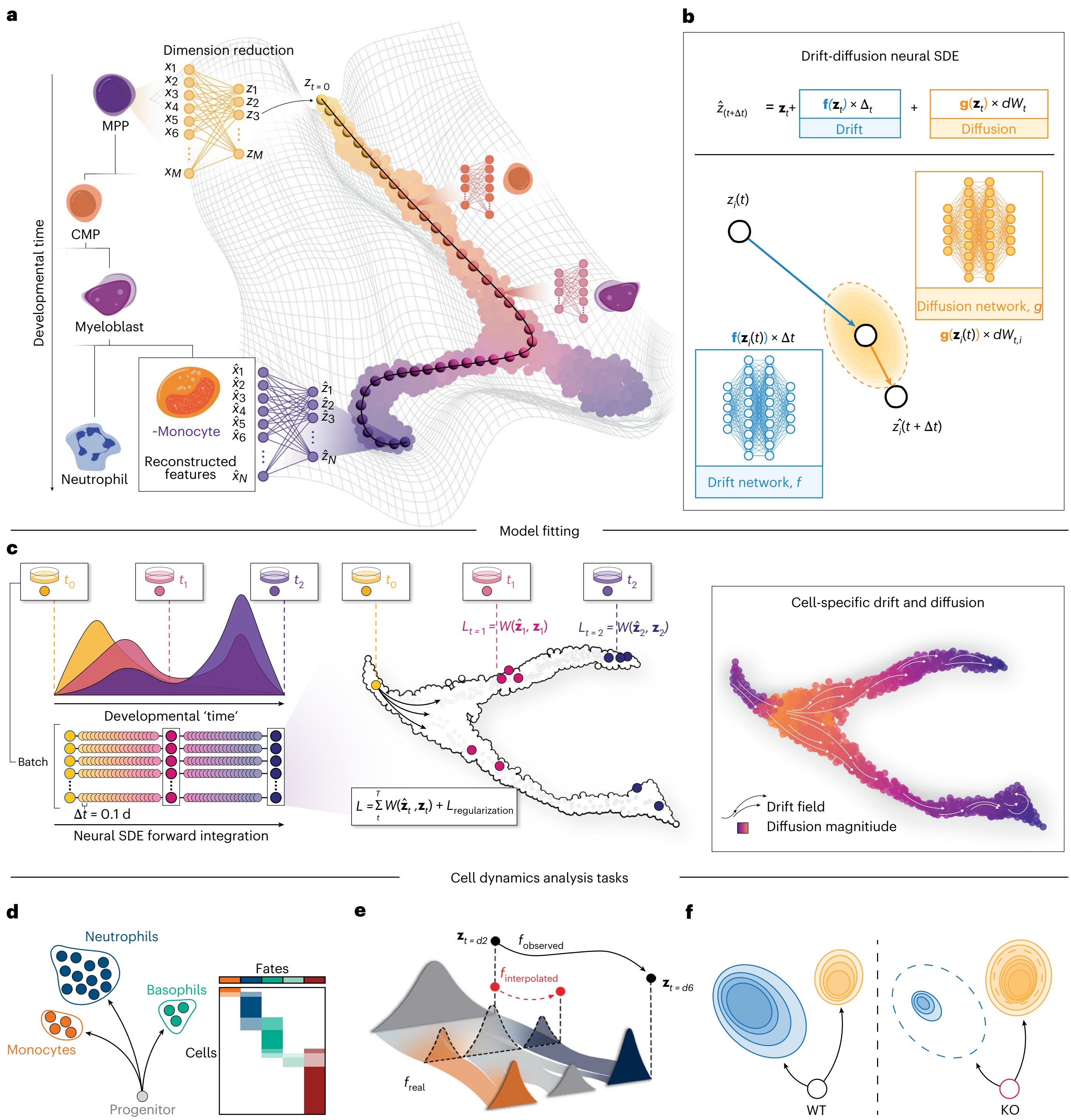


Fig. 1 | scDiffEq algorithm overview and applications. a, Conceptual overview of modelling cellular dynamics in haematopoietic development. The developmental landscape shows multipotent progenitors (MPP) differentiating through common myeloid progenitors (CMP) into specialized cell types including neutrophils, monocytes, and myeloblasts. Dimension reduction techniques capture this process in lower-dimensional space with reconstructed features. **b**, Neural SDE framework for modelling drift-diffusion dynamics. The equation $\hat{\mathbf{z}}(t+\Delta t) = \mathbf{z}_t + \mathbf{f}(\mathbf{z}_t) \times \Delta t + \mathbf{g}(\mathbf{z}_t) \times \Delta W(t)$ combines deterministic drift (blue, modelled by neural network, f) and stochastic diffusion (orange, modelled by neural network, g) terms to predict cell state changes over time. **c**, scDiffEq training schematic showing the optimization process. Cells are sampled in batches and evolved through developmental 'time' using the neural SDE. The model minimizes Wasserstein distance between generated ($\hat{\mathbf{z}}$) and observed ($\mathbf{z}$) cell distributions at multiple time points ($L_{t=1}, L_{t=2}$), with the total loss calculated as the sum of Wasserstein distances across all time points combined with a regularization loss. During training, a drift vector field and diffusion magnitude are learned. **d**, Task 1: fate prediction from multipotent progenitors to determine the probability of differentiation into specified fates (for example, neutrophils, basophils, monocytes). **e**, Task 2: temporal interpolation to reconstruct cell states at unobserved intermediate time points between observed time points. The true, unobserved function is f_{real}. The model is fit to f_{observed} with cell states, $\mathbf{z}_{t=d2}$ and $\mathbf{z}_{t=d6}$ and is evaluated on its ability to construct the output of $f_{\mathrm{interpolated}}$ at the unseen time point. **f**, Task 3: in silico perturbation analysis comparing wild-type (WT) with knockout (KO) conditions to predict how genetic perturbations affect cell fate trajectories and population dynamics.

https://doi.org/10.1038/s42256-025-01150-3

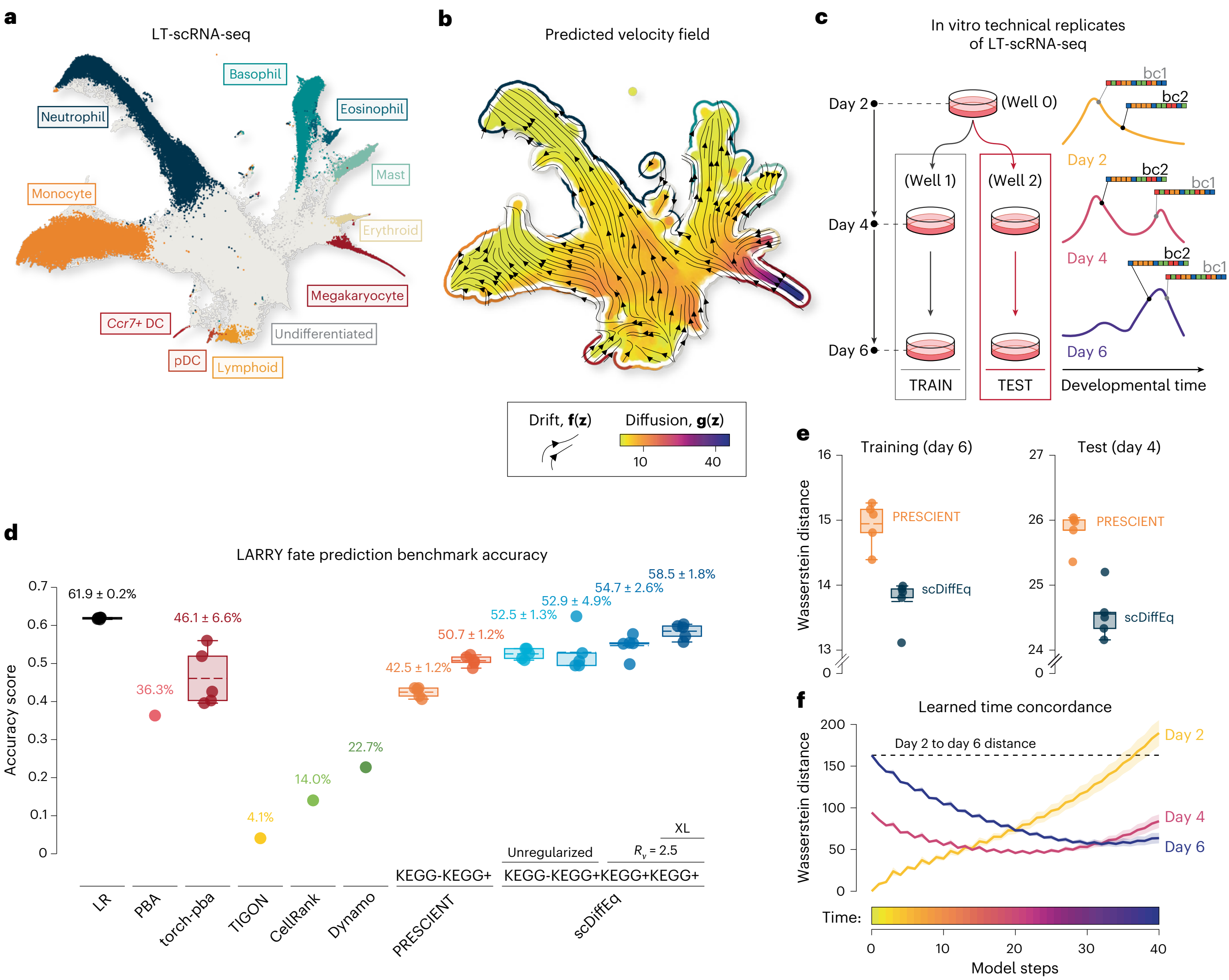


Fig. 2 | Benchmarking scDiffEq performance using lineage-traced haematopoietic development data. a, UMAP visualization of the LARRY scRNA-seq dataset (130,887 cells) coloured by cell type annotations. **b**, scDiffEq velocity field decomposition displayed as a UMAP stream plot. Vector field arrows represent learned drift dynamics (deterministic component), while the colour gradient shows diffusion magnitude (stochastic component, L2Norm) ranging from low (yellow) to high (purple) diffusion. **c**, Schematic overview of the lineage-tracing strategy and experimental setup of the in vitro LARRY dataset. **d**, Fate-prediction benchmark results (task 1) comparing accuracy scores across methods. Linear regression provides an upper baseline (61.9 ± 0.2%). scDiffEq achieves 58.5 ± 1.8% accuracy ($n = 5$) outperforming specialized single-cell methods including PRESCIENT (42.5–50.7%, $n = 5$), PBA (36.3%, $n = 1$), PyTorch-PBA (46.1 ± 6.6%, $n = 5$), Dynamo (26.3%, $n = 1$), CellRank (14.0%, $n = 1$) and TIGON (4.1%, $n = 1$). KEGG+ indicates use of growth-informed weights (precomputed by PRESCIENT) during the optimal transport loss calculation. R_v indicates the use of regularized velocity. XL indicates the more expressive version of scDiffEq uncovered through additional parameter tuning (Supplementary Fig. 9). Box plots display the mean (middle, dashed line), first and third quartiles (lower and upper box bounds), and whiskers extending to 1.5× the IQR; data points beyond the whiskers are shown as individual points. Data are presented as mean values ± standard deviation where indicated. n is the number of independent biological replicates, where each replicate represents an independent model training run with a different random initialization seed. **e**, Temporal interpolation performance (task 2) measured by Sinkhorn divergence between predicted and observed cell distributions. Models are trained on days 2 and 6, then evaluated on the withheld day 4 population. scDiffEq achieves a mean training distance of 13.74 (s.e.m.: 0.14, $n = 5$) and test distance of 24.56 (s.e.m.: 0.16, $n = 5$). PRESCIENT achieves a mean training distance of 14.94 (s.e.m.: 0.14, $n = 5$) and test distance of 25.85 (s.e.m.: 0.11, $n = 5$). Minimized Sinkhorn divergence indicates optimized prediction of intermediate, unseen cells at day 4. Box plots display the median (centre line), first and third quartiles (lower and upper box bounds), and whiskers extending to 1.5× IQR. $n = 5$ independent model training runs, each with a different random initialization seed. **f**, Model trajectory validation demonstrating temporal accuracy of learned dynamics. Wasserstein distance is plotted between scDiffEq-predicted cell populations and observed LARRY populations across simulation time steps ($dt = 0.1$ days). The model successfully recapitulates the temporal evolution of cell states: predicted populations show minimal distance to day 2 cells (yellow line) at early time steps, progressively approach day 4 cells (pink line) at intermediate steps and converge towards day 6 cells (purple line) at later steps. These temporal distance profiles demonstrate that scDiffEq captures authentic developmental progression. The horizontal dashed baseline (day 2 to day 6 distance) confirms that intermediate predictions represent biologically meaningful transitional states rather than artefacts of linear interpolation between terminal time points. Lines represent mean Wasserstein distance values, with shaded regions indicating ±1 standard deviation across $n = 5$ independent model training runs with different random initialization seeds.

regularization improved scDiffEq fate prediction by ~1.9%. Enforcing this ratio also brought scDiffEq's predictions close to the observed entropy (Supplementary Fig. 8c). We therefore applied a drift/diffusion ratio of 2.5 throughout our analyses (Supplementary Note 1 and Methods). Subsequently, hyperparameter optimization identified an optimal scDiffEq variant with a drift network of 2 2,048-unit hidden layers and a diffusion network of 2 256-unit hidden layers that further boosted performance by ~3.8% to 58.5 ± 1.8% ($n = 5$), designated as 'XL' (Fig. 2d). scDiffEq's performance represents a ~7.8% improvement over state of the art models in cell fate prediction.

Beyond majority fate accuracy, we evaluated the predicted probability distribution against the true fate distribution using negative cross-entropy. scDiffEq demonstrated superior distributional alignment compared with other methods, particularly for multipotent progenitors with complex fate outcomes (Supplementary Fig. 10). While majority fate-prediction accuracy enables comparison with previous benchmarks, future work should develop metrics that quantitatively assess how well the entire predicted fate distribution matches ground truth.

While the raw accuracy gain of scDiffEq over PRESCIENT (58.5 ± 1.8% versus 50.7 ± 1.2%, Fig. 2d) appears modest, it understates scDiffEq's qualitative advantages. By modelling state-dependent diffusion, scDiffEq better captures rare fates and produces more balanced predictions across all outcomes. Unlike PRESCIENT's largely discrete outputs, scDiffEq predicts fate distributions in varying proportions (Supplementary Fig. 11).

Ablation of cell-specific diffusion demonstrates that diffusion is critical for accurate fate trajectory prediction. When comparing scDiffEq with a diffusion-ablated version using isotropic, homogeneous Gaussian noise, we observed an 11.1% decrease in prediction accuracy and a ~3.4-fold increase in cross-entropy (Supplementary Fig. 12a,b). While scDiffEq predicts diverse fates, the diffusion-ablated model exclusively predicts single-fate outcomes, failing to capture multipotent developmental trajectories (Supplementary Fig. 12c–g). scDiffEq generated trajectories that effectively captured multi-fate developmental paths with continuous coverage across developmental space. In contrast, the diffusion-ablated model produced oversimplified, strictly mono-fate trajectories that lack fine-resolution state transitions. These findings underscore the importance of modelling cell state-specific diffusion for accurately representing cell fate decisions in developmental systems.

We found that dataset size contributes substantially to fate-prediction performance with 30,000 cells across three time points approaching full-dataset accuracy (Supplementary Note 2 and Supplementary Fig. 13).

While fate prediction illuminates state–fate relationships, discriminative models cannot recapitulate entire cell trajectories. Both scDiffEq and PRESCIENT enable prediction of intermediate, unobserved cell states. In task 2, following ref. 15, we benchmarked interpolation of cells from a withheld time point[15]. Models were trained on day 2 and day 6 cells, withholding day 4 cells for testing (Fig. 1e). Reconstruction was measured using Sinkhorn divergence. Across five seeds, PRESCIENT achieved a mean training distance of 14.94 (standard error of the mean (s.e.m.): 0.14) and test distance of 25.85 (s.e.m.: 0.11). scDiffEq achieved 13.74 (s.e.m.: 0.14) and 24.56 (s.e.m.: 0.16) (Fig. 2e). This performance was achieved on removing the potential gradient constraint from the drift function and scaling to a 32 million parameter variant of scDiffEq. To interrogate whether this improvement results purely from increased model capacity, we increased the capacity of PRESCIENT. However, this does not result in consistently improved performance (Supplementary Fig. 14), supporting the notion that state-dependent diffusion, rather than model expressivity alone, improves performance.

We note a distinct advantage in the approach taken by scDiffeq and PRESCIENT to learn and subsequently simulate a latent time, from coarse, real-time measurements of cells. Using a sample model prediction from scDiffEq, we demonstrate that, over the course of a simulated population, cells beginning with zero error from their sampled day 2 population move away from that population and eventually minimize their distance to the observed day 4 and day 6 populations at increments of 0.1 d (Fig. 2f).

Model predictions were corroborated using in silico perturbations (task 3). One research group introduced a framework for predicting the outcome of gene perturbations in progenitor cells based on learned cell dynamics[15]. Inspired by this and similar frameworks such as CellOT, we asked whether scDiffEq could simulate expected biological responses under perturbed conditions[15,34]. We applied in silico perturbations to an ensemble of TFs involved in granulopoiesis and neutrophil development: *Lmo4*, *Dach1*, *Klf4* and *Cebpe*[35–38]. We applied z-score perturbations to 200 day 2 progenitor cells and simulated trajectories (Methods). Comparing these with unperturbed control simulations, we qualitatively observed a shift in population density from monocytes towards neutrophils when the TF ensemble was overexpressed at $z = 10$ (Fig. 3a,b). Systematic perturbation analysis revealed dose-dependent effects: TF overexpression increased neutrophil fraction from 30% to 48% while decreasing monocytes from 28% to 14%, with knockdowns showing inverse effects (Fig. 3c,d). Comparatively, linear regression failed to capture dose-dependent perturbation responses, demonstrating that scDiffEq's explicit modelling of cellular dynamics is essential for biologically interpretable predictions (Supplementary Note 3 and Supplementary Fig. 15).

Many functional genomics screens probe gene regulatory components and identify candidate regulators[39]. To explore scDiffEq's utility for in silico hypothesis generation, we tested its potential as a complementary tool alongside experimental approaches to study gene regulation. GMPs represent a key branch point in haematopoiesis, giving rise to granulocyte lineages (for example, neutrophils, eosinophils, basophils) and agranulocyte lineages (for example, lymphocytes, monocytes). We investigated this developmental fork through an in silico fate perturbation screen of the LARRY dataset using scDiffEq (Fig. 3e–g).

To assess whether perturbation predictions degrade with distance from the training distribution, we examined the relationship between the distance imparted by a perturbation (in PCA space) and prediction accuracy. We used naturally occurring differential expression in fate-committed progenitors as a proxy for accuracy. We found PCA-transformed perturbation distances were weakly but significantly correlated with accuracy (neutrophils: $r = 0.16$, $P = 5.1 \times 10^{-3}$; monocytes: $r = 0.20$, $P = 2.0 \times 10^{-3}$; Supplementary Fig. 16a), suggesting scDiffEq's perturbation predictions remain robust within natural expression variation ranges. Even strong perturbations (z-scores of ±10) caused minimal displacement from the natural differentiation landscape, instead inducing dose-dependent progenitor redistributions towards or away from specific fates (Supplementary Fig. 16b–d).

Genome-wide in silico perturbation screens across five model replicates identified fate-specific regulators using meta-analysis (Methods). For each fate (monocyte, neutrophil, basophil), we identified genes that consistently showed statistically significant effects. A clustered heatmap of these effect sizes highlights coordinated groups of known regulators mediating GMP differentiation (Supplementary Fig. 17). We highlight key marker genes and TFs associated with each fate. For example, simulated *Gata2* overexpression strongly enriches basophils while depleting monocytes whereas *Gfi1* overexpression promotes neutrophil-fated cells at the expense of monocytes (Fig. 3e–g).

Housekeeping gene controls showed minimal effects, validating perturbation specificity[40,41]. Neutrophil commitment increased following *Gfi1* perturbation, with a statistically significant effect compared with controls. Suppressors of neutrophil fate, *Klf4* and *Irf8*, led to significantly reduced neutrophil production. Monocyte fate was promoted by *Csf1r*, *Klf4*, *Cebpb* and *Irf8*, and suppressed by *Gfi1* and *Cebpa*. Basophil formation increased with *Gata2* and *Cebpa* perturbation, and decreased with *Csf1r*, *Irf8*, *Gfi1* and *Csf3r*. These results confirm that scDiffEq distinguishes functional regulatory perturbations from non-specific effects (Supplementary Fig. 18), with known haematopoietic regulators showing significantly stronger responses ($P < 0.05$), and effect sizes 1.9–23.1 times greater than housekeeping controls. Perturbation predictions correlated strongly with natural expression patterns ($r = 0.67$–0.76, $P < 1\times10^{-33}$), validating biological relevance (Supplementary Fig. 19).

Several marker genes conferred strong fate biases when perturbed (Fig. 3e–g). While perturbing TFs in plastic cells is expected to alter fate, perturbing fate-associated markers (for example, *Mpo*, *Elane*, *Hbb*) is less canonical. This may reflect a mechanistic limitation of optimal transport-based models: perturbing a progenitor to resemble a fate may reduce its distance to that fate in PCA space, biasing its trajectory within the learned transport plan.

scDiffEq successfully transferred to human haematopoiesis data (1,947 cells), identifying known regulators and reproducing dose-dependent *SPI1* effects on monocyte fate[9,42] (Fig. 3h–l and Supplementary Fig. 20). As before, we caution that overexpression of late-stage markers in early progenitors is unlikely to drive corresponding fate outcomes. For example, simulated overexpression of *HBB* and *HBD* ($z = +10$) produced strong $\log_2$ fold-change effect sizes of 0.68 ($P = 6.3 \times 10^{-68}$) and 0.70 ($P = 6.7 \times 10^{-65}$), respectively. While these genes are expressed later in megakaryocyte-erythroid progenitor development, they are unlikely to act as master regulators when overexpressed in haematopoietic stem cells.

Contemporary methods including PRESCIENT, PI-SDE and TIGON require multi-time-point datasets[15,17,25]. We have applied scDiffEq to the LARRY dataset (three time points, lineage-traced) and human haematopoiesis (two time points). We next extended scDiffEq to a widely used pancreatic endocrinogenesis dataset (E15.5, 3,696 cells, 3′ 10X scRNA-seq) captured in a single snapshot[43] (Fig. 3m and Supplementary Fig. 21a). We assigned developmental times using CytoTRACE and applied scDiffEq to map drift-diffusion dynamics[44] (Fig. 3m and Supplementary Fig. 21). While uniform manifold approximation and projection (UMAP)-based velocity fields provide limited insight, we observed coherent trajectories, with ductal cells progressing towards endocrine fates. This demonstrates scDiffEq's compatibility with snapshot datasets, which constitute the majority of single-cell data.

To assess the impact of temporal resolution on scDiffEq, we trained models using varying numbers of discretized pseudotime points from the pancreatic endocrinogenesis dataset (Supplementary Fig. 22a). A model trained on only three of six available time points maintained comparable performance to the model trained on the full dataset, indicating that scDiffEq can learn cellular dynamics from sparse temporal data. Progressive degradation of pseudotime led to increasingly discordant velocity fields, underscoring scDiffEq's reliance on accurate pseudotime to infer meaningful developmental trajectories (Supplementary Fig. 22b).

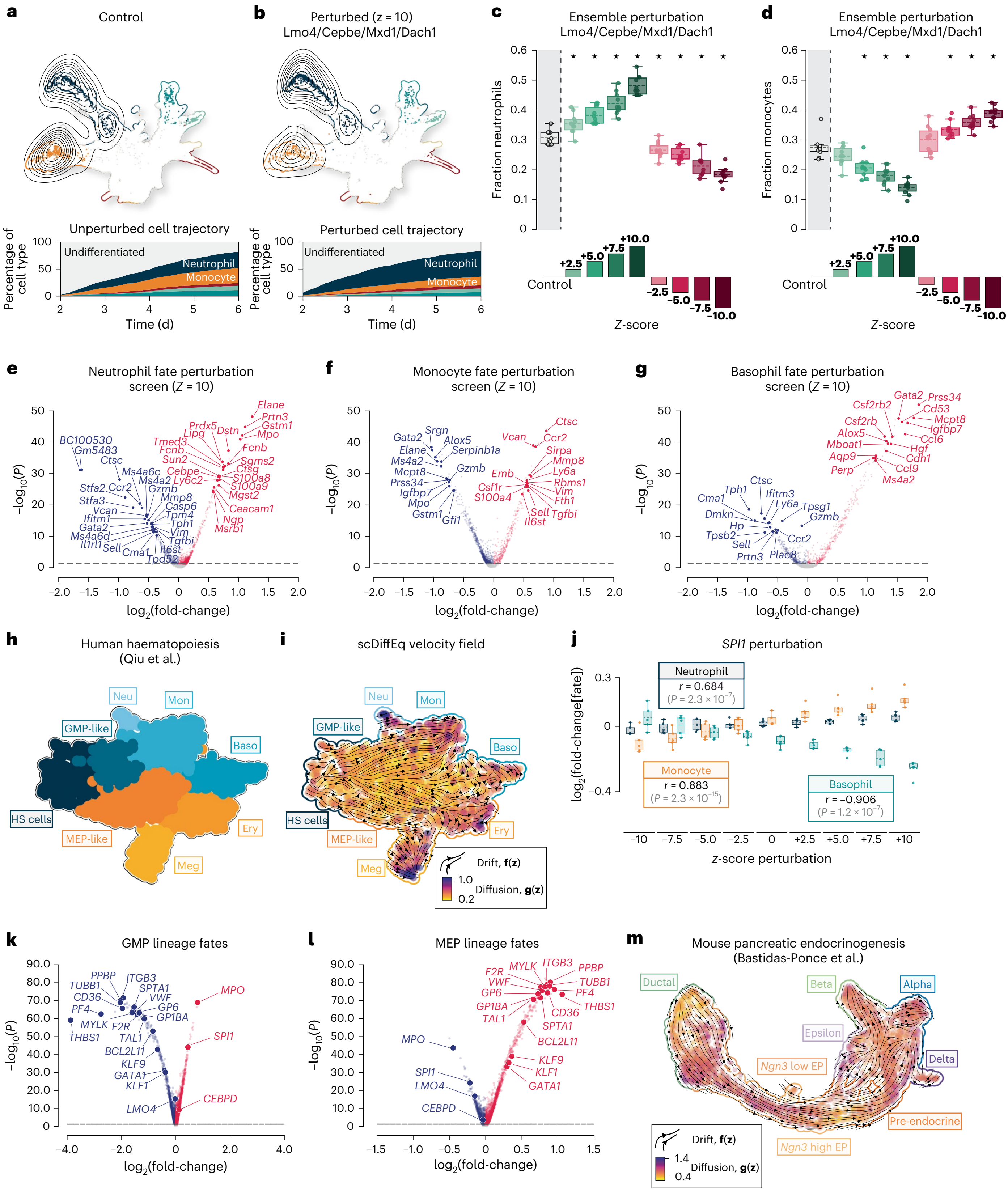
Decomposing the learned dynamics described by drift and diffusion

We have shown that scDiffEq (1) performs competitively on benchmark tasks, (2) serves as a valuable tool for modelling dynamics towards cell

Fig. 3 | In silico perturbation analysis and generalization across datasets. **a**,**b**, Validation of TF perturbation effects on haematopoietic fate decisions. UMAP projections show predicted cell distributions at day 6 under (**a**) control conditions and (**b**) overexpression of neutrophil-promoting TFs (Lmo4, Cebpe, Mxd1, Dach1 at a z-score of 10). Bottom panels show temporal evolution of cell type proportions, demonstrating a shift from balanced neutrophil or monocyte production to neutrophil-biased differentiation. **c**,**d**, Dose–response analysis of ensemble TF perturbation. Significance was tested by Welch's two-sided t-test (perturbed versus control, $P < 0.05$ indicated by stars). Data represent $n = 10$ replicate samplings (200 simulated trajectories each) from day 2 progenitors of the LARRY dataset (28,249 cells total). Box plots show mean (dashed line), quartiles (box) and 1.5× IQR whiskers. **c**, Neutrophil fraction increases with overexpression ($z = 2.5$–10.0: 35.4–48.2%) and decreases with knockdown ($z = -2.5$ to -10.0: 26.6–18.4%). Significant P values: +2.5: 1.4×10^{-3}; +5: 4×10^{-6}; +7.5: 2.4×10^{-7}; +10: 1.4×10^{-10}; −2.5: 1.5×10^{-2}; −5: 8.4×10^{-4}; −7.5: 1.0×10^{-5}; −10: 2.8×10^{-8}. **d**, Monocyte fraction shows the inverse trend, decreasing with overexpression (20.6–13.9%) and increasing with knockdown (33.5–38.8%). Significant P values: +5: 3.3×10^{-4}; +7.5: 8.4×10^{-6}; +10: 7.6×10^{-8}; −5: 8.6×10^{-4}; −7.5: 3.0×10^{-5}; −10: 7.8×10^{-7}. **e**–**g**, Genome-wide in silico perturbation screens in mouse haematopoiesis. Volcano plots show $\log_2$(fold-change) versus $-\log_{10}(P)$ for genes affecting neutrophil (**e**), monocyte (**f**) and basophil (**g**) fate formation on overexpression ($z = +10$). Key regulators are highlighted: *Gfi1* promotes neutrophil differentiation, *Gata2* drives basophil formation while depleting monocytes, and multiple known lineage-specific TFs and markers show expected directional effects. Statistical significance for each gene was assessed using Welch's independent two-sided t-test comparing perturbed with unperturbed conditions ($n = 10$ simulation replicates per gene, each with 200 simulated cells). P values were corrected for multiple testing using the Benjamini–Hochberg procedure to control the FDR (<0.05). Meta-analysis across five independent model replicates ensures robust identification of significant perturbation effects. Each point represents a single gene; genes meeting the significance threshold (FDR-adjusted $P < 0.05$) across all five model replicates are highlighted. The $\log_2$(fold-change) represents the $\log_2$ ratio of mean fate fraction between perturbed and unperturbed conditions. **h**,**i**, Application to human haematopoiesis dataset. **h**, UMAP of human scNT-seq haematopoiesis data (1,947 cells) showing haematopoietic stem (HS) cells, GMP-like, MEP-like and mature cell types (neutrophils, monocytes, basophils, erythrocytes, megakaryocytes)[42]. **i**, scDiffEq-learned velocity field decomposition showing drift vectors and diffusion magnitude (colour map), demonstrating successful model transfer to human data. **j**, *SPI1* perturbation dose–response in human haematopoiesis. Neutrophil, monocyte and basophil fate formation show dose-dependent response to *SPI1* expression levels. In silico overexpression of *SPI1* promotes neutrophil and monocyte differentiation while inhibiting basophil differentiation; knockdown inhibits neutrophil and monocyte differentiation while promoting basophil differentiation. Data represent $n = 5$ independent replicate samplings, where each replicate consists of 200 simulated cell trajectories generated from progenitor cells randomly sampled from the human haematopoiesis dataset (309 cells total). Each data point represents the mean fate fraction from one independent simulation replicate. The Pearson correlation coefficient, r indicates the correlation of $\log_2$(fold-change[fate]) to in silico z-score perturbation of *SPI1*. Statistical significance for each fate was computed as a two-sided test of non-correlation (H_0: $r = 0$) with P values from the exact null distribution of r under bivariate normality as implemented in scipy.stats.pearsonr. No adjustment for multiple comparisons was applied as each dose level represents a prespecified independent hypothesis. Box plots display the mean (middle, dashed line), first and third quartiles (lower and upper box bounds) and whiskers extending to 1.5× IQR. **k**,**l**, Human haematopoiesis perturbation screens identify regulators of GMP lineage (**k**) and MEP lineage fate specification (**l**). Known regulators such as *SPI1*, *GATA1* and lineage markers are recovered, validating the approach across species. Volcano plots show $\log_2$(fold-change) versus $-\log_{10}(P)$ for genome-wide perturbation effects ($z = +10$). Statistical significance for each gene was assessed using Welch's independent two-sided t-test comparing perturbed with unperturbed conditions ($n = 10$ simulation replicates per gene, each with 200 simulated cells). P values were corrected for multiple testing using the Benjamini–Hochberg procedure to control the FDR (<0.05). Meta-analysis across five independent model replicates ensures robust identification of significant perturbation effects. Each point represents a single gene; significant genes (FDR-adjusted $P < 0.05$) are colour-coded by their effect direction. **m**, Extension to single time-point data using mouse pancreatic endocrinogenesis[43]. scDiffEq successfully models developmental dynamics in a snapshot dataset (3,696 cells at E15.5) by using CytoTRACE-inferred developmental time. The velocity field shows ductal cells progressing towards endocrine fates (alpha, beta, delta cells), with drift vectors indicating differentiation trajectories and diffusion magnitude (colour map) revealing state-dependent stochasticity during pancreatic development.

https://doi.org/10.1038/s42256-025-01150-3

fate and (3) successfully been applied to three scRNA-seq datasets, including one without discrete time points. We next examined trajectories simulated by scDiffEq. We fit scDiffEq to the entire LARRY dataset over five independent seeds. We then simulated 2,000 trajectories from each of the 2,081 cell states observed in day 2 that are also heritably observed in later time points. We highlight three representative cases: a single-fate lineage (Fig. 4a), a bi-fated lineage (Fig. 4b) and a lineage with three or more fates (Fig. 4c). Prediction of single and multiple fates from the same model contrasts with methods that focus on parameterizing a more deterministic trajectory inference model.



We evaluated each measured cell of the LARRY dataset using scDiffEq's drift and diffusion neural networks to compute independent 50-dimension velocity vectors for both drift and diffusion, respectively. Each vector represents the instantaneous drift-specific or diffusion-specific 'velocity' of a state. We summarized drift and diffusion magnitudes as scalar L2 norms of these vectors, smoothed them via a nearest neighbour graph, and visualized the results with UMAP (Supplementary Fig. 23), observing cell-specific and group-specific fluctuations in both drift and diffusion.

We asked whether cell plasticity is correlated to drift and diffusion properties along developmental trajectories. We grouped lineage-traced trajectories by predicted fate multiplicity: single-fate, two-fate and three-or-more-fate outcomes. For each, we computed the L2 norm of drift and diffusion across five model replicates. We found that the maximum L2 norm of both the drift and diffusion increased monotonically with fate multiplicity: from mono-fate (drift: 53.1 ± 1.3, diffusion: 2.9 ± 0.5) to bi-fated (drift: 57.6 ± 0.8, diffusion: 3.6 ± 0.2) to ≥3 fates (drift: 60.7 ± 0.6, diffusion: 4.7 ± 0.2) (Fig. 4d).

We highlight the accurately simulated trajectory of a bi-fated progenitor from the LARRY dataset that produces monocytes and neutrophils in roughly equal proportion (Fig. 4e–j). We decomposed drift and diffusion along this trajectory and visualized the results with UMAP (Fig. 4e,f). While UMAP projections obscure the complexities of biological state space, we infer a state-dependent relationship between drift and diffusion along the neutrophil and monocyte developmental trajectories.

We analysed a representative trajectory predicting bi-fated neutrophil–monocyte outcomes. We reconstructed gene expression by inverting the predicted PCA state vector, calculated the L2 norm of drift and diffusion per cell and averaged these by fate at each time step ($dt = 0.1$ d). Correlating fate-conditional drift and/or diffusion with reconstructed TF gene expression revealed four distinct clusters through hierarchical clustering (Fig. 4g). These clusters reflected four distinct correlation patterns between TFs and dynamics: diffusion-correlated regulators (*Myc*, *Cebpa*), fate-specific factors (*Gfi1*, *Lmo4*), broadly anti-correlated genes (*Klf4*, *Dach1*) and neutrophil-specific patterns[35,45] (*Foxo1*, *Gata2*) (Fig. 4g).

Using scDiffEq, we model the time-dependent expression of *Spi1*, *Gfi1*, *Klf4* and *Irf8*, separately in monocyte-fated and neutrophil-fated trajectories (Fig. 4j–m). *Spi1* encodes the pioneer TF protein, PU.1 and is required for neutrophil development[35]. The GFI1 protein, encoded by *Gfi1*, is a transcriptional repressor in haematopoiesis. In early haematopoietic progenitors, *Gfi1* and *Spi1* regulate mouse haematopoiesis antagonistically through a protein–protein interaction[46]. Independent of fate, development beyond the GMP state is broadly dependent on *Spi1* expression[47]. Consistent with previous notions around *Gfi1* involvement in GMP development, we find sustained expression of *Gfi1* into neutrophil-annotated state space (Fig. 4i,k). *Spi1* expression is observed along each trajectory (Fig. 4h,j). Taken together, scDiffEq describes a high-resolution, dynamic portrait of *Spi1* and *Gfi1*, highlighting how their antagonistic interplay shapes haematopoiesis.

We benchmarked the computational requirements of scDiffEq against CellRank, Dynamo and PRESCIENT. scDiffEq was more computationally demanding than CellRank and Dynamo although scaled more favourably in terms of memory usage. PRESCIENT scaled similarly to scDiffEq (Supplementary Fig. 24).

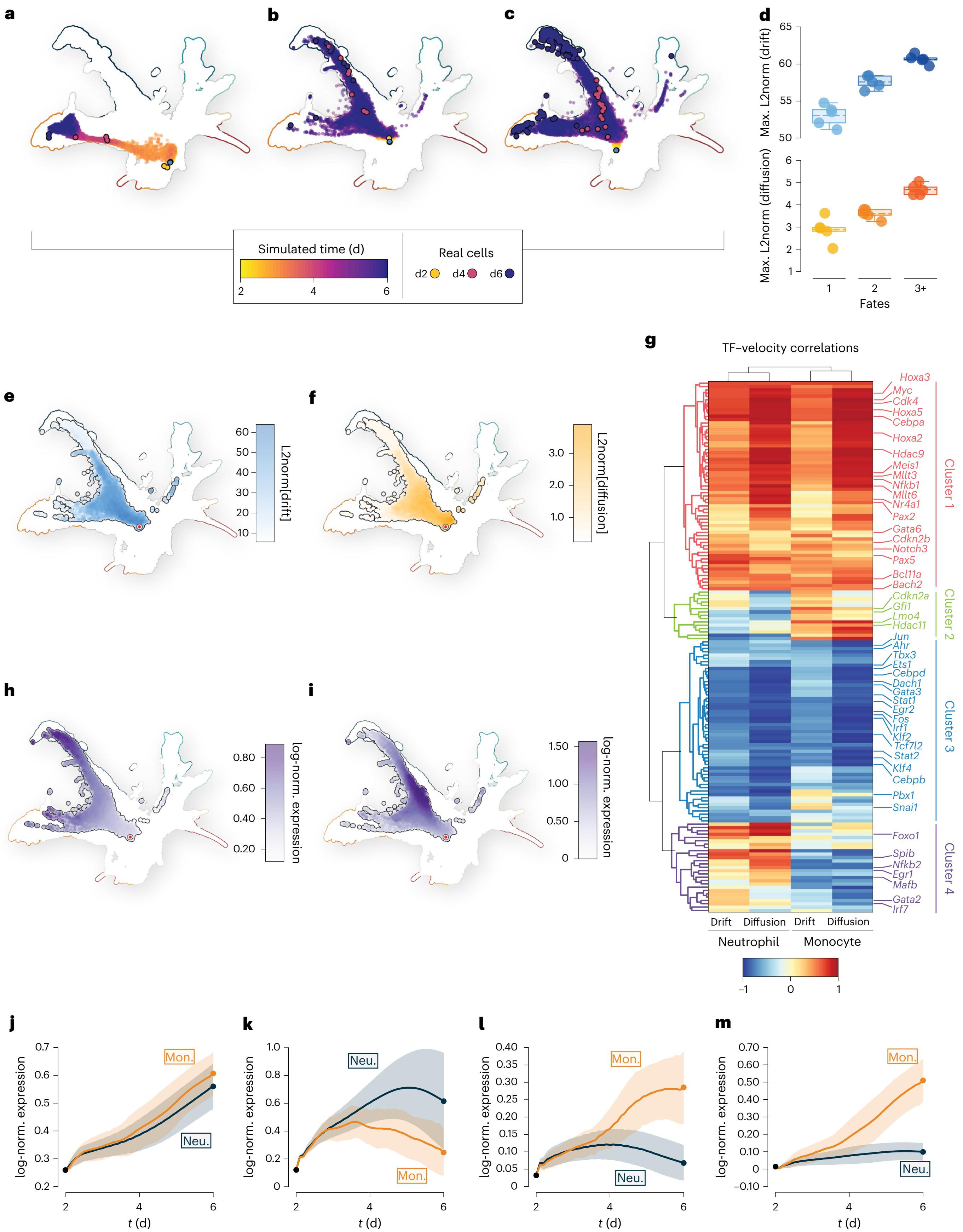
We profiled the computational scalability of scDiffEq across varying dataset sizes (1,000 to 1 million cells) and trajectory durations (2–9 days). Using only standard hardware, we found scDiffEq scales to datasets exceeding 1 million cells with near-linear scaling, approximated as $O(N^{0.88\pm0.002} \times T^{0.73\pm0.01})$ (Supplementary Fig. 25 and Methods). We also analysed training time and memory requirements for fate prediction (task 1) spanning across network complexities (Supplementary Figs. 26 and 27). Combined, these analyses provide the user with a robust reference dataset for tuning model size, batch size and dataset size independently to fit their computational constraints while preserving scDiffEq's favourable scaling properties.

Discussion

scDiffEq is a generative drift-diffusion framework based on neural SDEs for modelling single-cell dynamics. Benchmarking against the lineage-traced LARRY dataset shows that scDiffEq outperforms contemporary methods in reconstructing cell dynamics, measured by cell fate prediction (task 1) and unseen population reconstruction

Fig. 4 | Decomposition of cell-specific drift-diffusion dynamics and transcriptional regulation during lineage commitment. **a**–**c**, Simulated trajectory diversity from individual day 2 progenitor cells demonstrates scDiffEq's ability to model variable fate outcomes. UMAP projections show representative trajectories from a mono-fate progenitor producing primarily one cell type (**a**), a bi-fated progenitor generating both neutrophils and monocytes (**b**) and a multi-fate progenitor (**c**) (≥3 fates) with broad differentiation potential. Simulated cells are coloured by developmental time (yellow to purple gradient) and overlaid on the observed LARRY cell manifold. Real lineage-traced cells from the corresponding barcode families are shown as coloured dots (day 2: yellow, day 4: pink, day 6: purple). **d**, Quantitative relationship between fate multiplicity and dynamics magnitude. Both maximum drift (deterministic component) and diffusion (stochastic component) L2 norms increase monotonically with lineage plasticity: mono-fated (drift: 53.1 ± 1.3, diffusion: 2.9 ± 0.5), bi-fated (drift: 57.6 ± 0.8, diffusion: 3.6 ± 0.2) and multi-fated (drift: 60.7 ± 0.6, diffusion: 4.7 ± 0.2) progenitors. Box plots display the mean (dashed middle line), first and third quartiles (lower and upper box bounds) and whiskers extending to 1.5× IQR. Values are presented as mean ± standard deviation across $n = 5$ independent replicates, where each replicate is an scDiffEq model trained with different random initialization seeds on the lineage-traced LARRY dataset. Within each replicate, maximum L2 norms of drift and diffusion were computed across all lineage-traced trajectories grouped by experimentally observed fate multiplicity (mono-fated, bi-fated or multi-fated outcomes). The unit of study is the lineage-traced progenitor cell trajectory, with fate multiplicity determined from experimental lineage barcoding. **e**,**f**, State distribution of dynamics components along a representative bi-fated neutrophil–monocyte trajectory. **e**, Drift magnitude (L2 norm, 0–60 scale) shows deterministic velocity patterns, with higher values in transition regions. **f**, Diffusion magnitude (L2 norm, 0–3.0 scale) reveals state-dependent stochasticity, with elevated diffusion at decision points during lineage commitment. **g**, Hierarchical clustering of TF correlations with drift and diffusion components across neutrophil and monocyte trajectories. The heatmap identifies four distinct clusters: cluster 1 (red), factors strongly correlated with diffusion in both lineages (including *Myc*, *Cebpa*, *Meis1*); cluster 2 (green), factors with predominantly monocyte lineage-specific correlations (*Gfi1*, *Lmo4*); cluster 3 (blue), factors anti-correlated with both drift and diffusion (*Klf4*, *Dach1*, *Cebpd*) across both lineages; cluster 4 (purple), factors with predominantly neutrophil lineage-specific correlation patterns (*Foxo1*, *Gata2*, *Mafb*). **h**,**i**, Expression patterns of key regulatory factors. **h**, *Spi1* (encoding PU.1) shows broad expression across both neutrophil and monocyte trajectories, consistent with its role as a master regulator of myeloid development. **i**, *Gfi1* expression is predominantly restricted to neutrophil-fated trajectories, reflecting its specific role in neutrophil differentiation and antagonistic relationship with monocyte development. **j**–**m**, Temporal gene expression dynamics during fate-specific differentiation. Reconstructed expression profiles show *Spi1* maintaining high expression in both neutrophil and monocyte trajectories with slight lineage-specific differences (**j**), *Gfi1* exhibiting sustained expression specifically in neutrophil trajectories while remaining low in monocyte trajectories (**k**), *Irf8* showing preferential expression in monocyte trajectories (**l**) and *Klf4* displaying higher expression in monocyte compared with neutrophil trajectories (**m**). Solid lines represent mean expression trajectories, with shaded regions indicating one standard deviation across simulated cells. These patterns recapitulate known regulatory relationships in GMP differentiation. max., maximum; norm., normalized.

https://doi.org/10.1038/s42256-025-01150-3



(task 2). We also use scDiffEq to predict the effect of in silico perturbations (task 3). We applied scDiffEq across three scenarios: lineage-traced multi-time-point data, two-time-point data without lineage tracing and single-snapshot datasets typical of scRNA-seq, demonstrating broad applicability and establishing clear benchmarks for future studies.

Beyond methodological improvements, scDiffEq advances our ability to study cell dynamics by explicitly modelling both deterministic (drift) and stochastic (diffusion) components. By simulating differentiation from haematopoietic progenitors, we described how drift and diffusion fields vary across trajectories and terminal fates. These simulated trajectories recapitulate experimentally observed gene expression patterns in neutrophil and monocyte lineages. The model's ability to disentangle deterministic from stochastic contributions offers a quantitative framework for probing fate decisions, with potential to clarify their relative roles in development and disease. Notably, we observe distinct drift and diffusion patterns across trajectories, suggesting that cells actively modulate their susceptibility to stochastic state changes during differentiation.

Modelling state-dependent diffusion (versus constant diffusion) enables more accurate representation of the full spectrum of cell fates, particularly rarer and multi-fate outcomes. This approach better captures progenitor cells' true multipotent nature while providing new insights into how deterministic and stochastic processes vary across developmental trajectories, analyses impossible with previous methods.

Unlike descriptive methods based on RNA velocity or cell–cell similarity, scDiffEq is a fully generative model capable of simulating realistic cellular trajectories. This allows scDiffEq not only to reconstruct developmental paths, but also perform in silico perturbations and counterfactual simulation, positioning it as a scalable platform for hypothesis generation and mechanistic exploration.

The technical limitations of scDiffEq highlight valuable directions for future development. Relying on a precomputed PCA latent space may obscure gene–gene relationships, limiting scDiffEq's ability to fully model gene network interactions. Our in silico screen highlights markers such as *Mpo* and *Elane*, although in vitro overexpression in haematopoietic stem cells would be unlikely to produce the same effect. We posit this reflects a broader limitation of optimal transport models: by optimizing least-action paths in a low-dimensional space, they may miss finer details of development. Modelling dynamics in gene space may overcome this challenge.

Despite these limitations, scDiffEq demonstrates favourable near-linear scaling over dataset sizes up to 1 million cells and biological contexts spanning 10 time points over 9 days. Simulation-free flow-matching algorithms offer promising avenues for further scalability improvements[48,49]. Building on this scalable foundation, future directions include multimodal integration of chromatin accessibility and protein data to provide comprehensive insights into cell state transitions and their dysregulation in disease. scDiffEq is uniquely positioned to quantify each modality's contribution to fate determination. Technical refinements could involve alternative dimension reduction methods that better preserve gene–gene relationships and improved regularization for stochastic decision-making. Experimental validation through targeted perturbations and real-time cell tracking will be essential to confirm model predictions.

scDiffEq offers a flexible drift-diffusion framework for modelling single-cell dynamics, with potential applications extending to spatial transcriptomics and multiplexed perturbation assays. As a generalizable method for training neural SDEs, scDiffEq establishes a foundation for biologically informed neural differential equation-based models. scDiffEq represents an important methodological step towards developing the next generation of generative models for studying biological dynamics from single-cell data.

Methods

scDiffEq model framework

Drift-diffusion model of cell dynamics. We model cell dynamics using a drift-diffusion equation where $x \in \mathbb{R}^D$ represents a cell's position in gene expression space and $z \in \mathbb{R}^L$ denotes its corresponding low-dimensional representation (for example, 50 principal components). The cell state dynamics follow:

$$\Delta \mathbf{z}(t) = \mathbf{f}(\mathbf{z}(t))\,\Delta t + \mathbf{g}(\mathbf{z}(t))\,\Delta W_t \quad (1)$$

where $\mathbf{f}(\mathbf{z})$ and $\mathbf{g}(\mathbf{z})$ are the drift and diffusion vector fields defined on R^L, respectively. Here, t is an unobserved latent time. $\mathbf{f}(\mathbf{z}(t))$ produces a vector describing the drift at state $\mathbf{z}(t)$ scaled by Δt. ΔW_t represents an increment of Brownian motion over the time interval, Δt as a standard spherical Gaussian, with $\Delta W_{i,j}(t) \sim N(0,\Delta t)$. W is a standard Wiener process. For batched cells, $\mathbf{z}(t)$ is represented by 50 principal component vectors, ΔW_t is of shape: [cells, 1], which is broadcast to shape: [cells, 50, 1] for element-wise multiplication with the predicted the diffusion vector, $\mathbf{g}(\mathbf{z}(t))$. Forward integration yields future cell states:

$$\mathbf{z}(t_1) = \mathbf{z}(t_0) + \int_{t_0}^{t_1} \mathbf{f}(\mathbf{z}(t))\,dt + \int_{t_0}^{t_1} \mathbf{g}(\mathbf{z}(t))\,dW_t \quad (2)$$

Optionally, the drift field, $\mathbf{f}(\mathbf{z}(t))$ may be constrained to the negative gradient of a potential function:

$$\mathbf{f}(\mathbf{z}(t)) = -\nabla\psi(\mathbf{z}(t)) \quad (3)$$

This yields the alternative form of the drift-diffusion equation as:

$$\Delta \mathbf{z}(t) = -\nabla\psi(\mathbf{z}(t))\,\Delta t + \mathbf{g}(\mathbf{z}(t))\,\Delta W_t \quad (4)$$

Model training and implementation

We parameterized $\mathbf{f}(\mathbf{z}(t))$ and $\mathbf{g}(\mathbf{z}(t))$ as deep neural networks in a neural SDE to reflect the model described in equation (2). We used first-order Euler discretization via 'torchsde.sdeint' from Google Research with SDE type "ito" and noise type "general":

$$\mathbf{z}(t+\Delta t) \approx \mathbf{z}(t) + \mathbf{f}(\mathbf{z}(t))\,\Delta t + \mathbf{g}(\mathbf{z}(t))\,\Delta W_t \quad (5)$$

The model requires specification of initial time, t_{init} and corresponding cell states, $\mathbf{z}(t_{\mathrm{init}})$. Cells of batch size N are randomly sampled from $\mathbf{z}(t_{\mathrm{sim}})$and forward-integrated to produce predicted states, $\mathbf{z}(t+n\Delta t)$, where $n \in \{0,1,2,\ldots,F\}$ and $F\Delta t = T$ where T is the final time in the dataset[22–24]. Δt is a hyperparameter specified as 0.1 throughout this work.

Neural networks were implemented in PyTorch with the following architectures:

Drift network, f. Two hidden layers of 512 units each with LeakyReLU activation with no dropout. During a subsequent hyperparameter optimization screen, we found two hidden layers of 2,048 parameters to be ideal for fate prediction (task 1). For time-point interpolation (task 2), 2 hidden layers of 4,000 units were used.

Diffusion network, *g*. Two hidden layers of 32 units each with LeakyReLU activation with no dropout. During a subsequent hyperparameter optimization screen, we found two hidden layers of 256 parameters to be ideal for fate prediction (task 1).

Optimization. Models were trained using the RSMProp optimizer with a learning rate of 0.0001. The StepLR learning rate schedular was implemented with step size of 1,500. A 90%/10% training/validation split was used. SDE integration used Ito calculus with $\Delta t = 1$. Only a single Brownian motion dimension was used throughout. For fate prediction (task 1), we used stochastic weight averaging throughout training.

Python implementation. scDiffEq is available as a Python package ('pip install scdiffeq') and compatible with the scverse ecosystem[50]. Training follows this pseudocode:

'''pseudocode

```
for epoch in range(train_epochs):
  for batch in training_data:
    time, X_true = batch
    X0=X_true[0] # first time point
    X_pred = SDEModel(X0, time)
    loss_sinkhorn = compute_sinkhorn_loss(X_pred,
    X_true)
    loss_velo_reg = compute_velocity_ratio_
    regularization(X_pred, velo_target)
    loss = loss_sinkhorn + loss_velo_reg
    update_parameters(loss)
```

'''

Practical learning of neural differential equations with scDiffEq. scDiffEq accepts cell transcriptional input of any dimension from scRNA-seq data, with or without time points. Contemporary methods, including PRESCIENT, typically apply PCA to reduce dimensionality[15]. For consistency, we used the first 50 principal components of a z-scored cell-by-gene expression matrix. scDiffEq requires annotation of an initial position to solve an initial value problem and fit the neural SDE that models dynamics over the cell manifold. scDiffEq computes the Wasserstein distance (approximated as Sinkhorn divergence) between sampled observed cells and model predictions.

Loss function

Sinkhorn divergence. As previously implemented, we used the unbiased, entropically regularized Wasserstein distance, computed via Sinkhorn divergence as a loss metric in the reconstruction of cell populations[6,15,51]. Models that aim to reconstruct cell populations require a metric to compare simulated cell populations with observed cell populations. The Wasserstein distance, also known as the earth mover's distance, quantifies the minimal cost of transporting mass from one probability distribution to another. The Wasserstein distance thus provides a useful measure of the distance between two probability distributions. Specifically, we use the Wasserstein distance, computed via Sinkhorn divergence to quantify the distance between simulated cell populations, υ_{sim} and corresponding observed cell populations, μ_{obs}. μ_{obs} and υ_{sim} are treated as probability measures on metric space, X. We may write the p-Wasserstein distance as:

$$W_p(\mu_{\text{obs}}, \upsilon_{\text{sim}}) = \left(\int_{X\times X} d(x,y)^p \, d\gamma(x,y)\right)^{\frac{1}{p}} \tag{6}$$

where $\prod(\mu_{\text{obs}}, \upsilon_{\text{sim}})$ denotes the set of all joint distributions, γ with marginals μ_{obs} and υ_{sim}. $d(x,y)$ is a distance metric on X wherein x and y represent points in X or the starting and ending points of mass transport in the optimal transport framework. As written, p-Wasserstein distance considers all possible p-metrics where p is a non-negative integer or real number that describes the type of norm used to measure the distance between points in support of their corresponding distributions. $p = 1$ describes the earth mover's distance and is more sensitive to the spread of mass distribution while $p = 2$, more frequently used in machine learning, provides a Euclidean-like measure of distance. Due to exponentiation, higher values of p generally lead to a distance metric more sensitive to outliers and large deviations between distributions. Here, we use a $p = 2$ Wasserstein distance for its Euclidean-like distance measure properties.

As above, allow $\mathbf{z} \in R^L$ to be a vector describing a cell's position in a low-dimension representation of gene expression space (for example, 50 principal components). For some discrete cell population, $z_{\text{obs}} = \{\mathbf{z}_1^{\text{obs}}, \mathbf{z}_2^{\text{obs}}, \ldots, \mathbf{z}_N^{\text{obs}}\} \in R^L$ and a corresponding, model-simulated population $z_{\text{sim}} = \{\mathbf{z}_1^{\text{sim}}, \mathbf{z}_2^{\text{sim}}, \ldots, \mathbf{z}_N^{\text{sim}}\} \in R^L$, we may describe the squared Euclidean distance between vectors of z_{obs} and z_{sim} as $\| \mathbf{z}_i^{\text{obs}} - \mathbf{z}_j^{\text{sim}} \|_2^2$. For vector $\mathbf{z} \in R^L$, this can be written as:

$$\sqrt{\sum_{k=1}^{D} (x_k - y_k)^2} \tag{7}$$

We further define the two discrete probability measures, $\mu_{\text{obs}} = (z_{\text{obs}}, \mathbf{w}_{\text{obs}}) \in R^L$ and $\upsilon_{\text{sim}} = (z_{\text{sim}}, \mathbf{w}_{\text{sim}}) \in R^L$ where $\mathbf{w}_{\text{obs}}$ and $\mathbf{w}_{\text{sim}}$ are non-negative weight vectors on the standard simplex, Δ^D such that observed cells' weights, $\sum_{i=1}^{n} w_i^{\text{obs}} = 1$ and $w_i^{\text{obs}} \geq 0 \forall i \in \{0, 1, \ldots, n\}$. The entropically regularized Wasserstein distance between μ_{obs} and υ_{sim}, with a squared Euclidean cost can be written:

$$W_\lambda(\mu_{\text{obs}}, \upsilon_{\text{sim}}) = \langle P, M \rangle + \lambda \text{KL}(P || \mu_{\text{obs}} \otimes \upsilon_{\text{sim}}) \tag{8}$$

where P is the transport plan in the set of all transport plans $U(\mu_{\text{obs}}, \upsilon_{\text{sim}})$ that maps μ_{obs} to υ_{sim} and M is the cost matrix representing the cost of transporting mass between μ_{obs} and υ_{sim}. The Frobenius inner product of the transport plan matrix P and the cost matrix, M, $\langle P, M \rangle$ describing the cost of the transport plan, is computed as the sum of element-wise products:

$$\langle P, M \rangle = \sum_{i=1}^{n} \sum_{j=1}^{m} P_{i,j} M_{i,j} \tag{9}$$

$\langle P, M \rangle$ thus represents the total cost of the transport plan matrix, $P_{i,j}$ according to the cost matrix, $M_{i,j}$ wherein we describe the cost of transporting mass from the ith source to the jth target. KL denotes the Kullback-Leibler divergence between the transport plan, P and $\boldsymbol{\mu}_{\text{obs}} \otimes \boldsymbol{\upsilon}_{\text{sim}}$. $\boldsymbol{\mu}_{\text{obs}}$ is an n-dimension vector and $\boldsymbol{\upsilon}_{\text{sim}}$ is an m-dimension vector such that $\mu_{\text{obs}} \otimes \upsilon_{\text{sim}}$ is an $n \times m$ matrix so that the ith row and jth column of that matrix represent the joint probabilities $\mu_{\text{obs},i} \upsilon_{\text{sim},j}$, assuming the positions in μ_{obs} and υ_{sim} are independent. λ is a scalar regularization term applied to the KL divergence, balancing the trade-off between the cost of transport and entropy in the solution.

The marginals of the transport plan must be equal to the measures μ_{obs} and υ_{sim}. Specifically, the row marginals of $P_{i,j}$, the sum of each row, i of matrix $P_{i,j}$ represents the total mass being transported from the ith point of the observed distribution, μ_{obs} to the jth point in the corresponding simulated distribution, υ_{sim}. This sum must equal w_i^{obs} where w_i^{obs} is the weight, the probability associated with the ith observed point:

$$\sum_{j=1}^{m} P_{i,j} = w_i^{obs} \forall i \tag{10}$$

Similarly, the column marginals of $P_{i,j}$, the sum of each column, j of matrix $P_{i,j}$ represents the total mass being transported to the jth point in the simulated distribution, υ_{sim}. This sum must equal w_j^{sim} where w_j^{sim} is the weight: the probability associated with the jth simulated point:

$$\sum_{i=1}^{n} P_{i,j} = w_j^{\text{sim}} \forall j \tag{11}$$

Having defined the regularized Wasserstein distance and constrained the marginal properties of the masses transported,

https://doi.org/10.1038/s42256-025-01150-3

the Sinkhorn divergence between two distributions μ_{obs} and υ_{sim} may be written:

$$S_\lambda(\mu_{\text{obs}}, \upsilon_{\text{sim}}) = W_\lambda(\mu_{\text{obs}}, \upsilon_{\text{sim}}) - \frac{1}{2}(W_\lambda(\mu_{\text{obs}}, \mu_{\text{obs}}) + W_\lambda(\upsilon_{\text{sim}}, \upsilon_{\text{sim}})) \quad (12)$$

By including self-transport terms, $W_\lambda(\mu_{\text{obs}}, \mu_{\text{obs}})$ and $W_\lambda(\upsilon_{\text{sim}}, \upsilon_{\text{sim}})$, the Sinkhorn divergence is equal to zero when $\mu_{\text{obs}} = \upsilon_{\text{sim}}$. There are infinite transport plans in the set of all transport plans $U(\mu_{\text{obs}}, \upsilon_{\text{sim}})$. Fortunately, the solution to finding the optimal transport plan, $P_{i,j}$ is a convex optimization problem that may be solved efficiently using a previously implemented graphical processing unit (GPU)-based solver[51].

In scDiffEq and in similar methods, such as PRESCIENT, each cell type is modelled as a particle in a distribution. The Sinkhorn divergence regularizes the optimal transport problem, enabling efficient and scalable computation of Wasserstein distances between observed and predicted distributions of cells.

Regularization through drift/diffusion ratio

We define the ratio of velocity as:

$$R_{\mathbf{v}} = \frac{\|\mathbf{v}_f\|_2}{\|\mathbf{v}_g\|_2} \quad (13)$$

where the numerator describes the Euclidean norm of the drift vector produced at a given cell state and the denominator describes the Euclidean norm of the corresponding diffusion vector. To impose pressure on the model to converge towards maintenance of a desired ratio of velocity, $R_{\upsilon,\text{target}}$ we first compute R_υ for a batch of N cells, $R_{\upsilon,\text{batch}}$. We then compute the mean square error over the batch, between the target, $R_{\upsilon,\text{target}}$ the predicted batch ratio, $R_{\upsilon,\text{batch}}$. This error is subsequently multiplied by a constant value, E, which serves as a regularizing scalar by which one may differentially enforce effect of regularizing the ratio of velocity, giving:

$$L_{R_{\mathbf{v}}} = \frac{1}{N}\sum_{i=1}^{N}(R_{\mathbf{v},\text{target}} - r_i)^2 \quad (14)$$

where r_i is the ratio of velocity computed for a cell in $R_{\upsilon,\text{batch}}$. Throughout this work, E was set to 100 (the hyperparameter was screened as shown in Supplementary Fig. 9). $L_{R_{\mathbf{v}}}$ may be added to the loss derived from the Sinkhorn divergence, L_S, to arrive at the total loss, L_{total} over which backpropagation is performed.

$$L_{\text{total}} = L_S + L_{R_\upsilon} \quad (15)$$

Datasets and preprocessing

LARRY. The dataset comprises 130,887 cell profiles: 49,302 (37.7%) were successfully transduced with one of 5,864 lineage barcodes. Cell counts were 28,249 (day 2), 48,498 (day 4) and 54,140 (day 6), with barcode detection in 16.4%, 30.9% and 54.8% of cells, respectively. On day 2, 4,638 barcoded cells spanned 2,672 unique barcodes; on day 4, 14,985 barcoded cells spanned 4,101 barcodes and on day 6, 29,679 barcoded cells spanned 3,956 barcodes. The most abundant barcodes occupied 8, 56 and 157 cells at days 2, 4 and 6, respectively (0.17%, 0.38% and 0.53% of barcoded cells).

Data matrices and metadata were downloaded from GitHub (https://github.com/AllonKleinLab/paper-data/tree/master/Lineage_tracing_on_transcriptional_landscapes_links_state_to_fate_during_differentiation, commit: af842ce). Following the procedure from PRESCIENT, we filtered non-highly variable genes, regressed out genes associated with a list of cell cycle genes[15]. For downstream analyses, we hand-selected several TFs and marker genes that were described in the original publication of the dataset or used in the analyses describing PRESCIENT for re-inclusion before subsequent processing[15,26]. We then scaled the log-normalized expression counts and performed PCA using scikit-learn.

Human haematopoiesis. The dataset, including the final list of filtered cells was provided by the authors of ref. 9. Following the procedure described in ref. 9, cells were filtered according to those that were published (to facilitate comparison), and the dataset was preprocessed using the Scanpy function, sc.pp.recipe_seurat[9,50,52]. The preprocessed data matrix was centred by subtracting the mean and normalizing to variance one, using scikit-learn. PCA to capture the first 50 principal components was performed using scikit-learn.

Pancreatic endocrinogenesis. The dataset was downloaded as an AnnData[50] object from GitHub (https://github.com/theislab/scvelo_notebooks/raw/master/data/Pancreas/endocrinogenesis_day15.h5ad, commit: 2425e7f). Cells with fewer than 200 genes expressed, and genes expressed in fewer than three cells were filtered from the dataset. Expression counts were log-transformed. Highly variable gene selection was performed. The filtered set of log-normalized genes were standardized by subtracting the mean and normalizing to variance one, using scikit-learn. PCA to capture the first 50 principal components was performed using scikit-learn.

Benchmark task 1: fate prediction

scDiffEq. We fit scDiffEq over 250 epochs, exposing the model to cells from well 0 (day 2) and well 1 (day 4 and day 6), using 90% of that data for training and 10% for validation, reshuffling that split at every epoch. Evaluation was performed by predicting the clonal fate bias of cells in well 2 (day 4 and day 6) from the corresponding clonal progenitors in well 0 (day 2).

PRESCIENT. We fit PRESCIENT, using the recommended parameters as previously described in ref. 15. Briefly, using cells from well 0 (day 2) and well 1 (day 4 and day 6), PRESCIENT was trained over 2,500 epochs. Evaluation was performed by predicting the clonal fate bias of cells in well 2 (day 4 and day 6) from the corresponding clonal progenitors in well 0 (day 2). Predicted cell states at day 6 were labelled using the prefit nearest neighbour classifier.

PBA. To predict fate bias with PBA, we used the implementation described in its original publication. Following the procedure described in ref. 13, we independently sampled 20,000 cells from the training set over five seeds[13,15,26]. PBA requires designating 'source' and 'sink' points in the cell manifold. Undifferentiated cells were designated as the source. To designate the sink points, we first precomputed the potential values for each cell in the training set. For each cell fate, the cell with the minimum value of potential and its 20 nearest neighbours were designated as sink points. $S = 10$ was used as the corresponding S parameter for each sink point cell. Undifferentiated cells were assigned $R = 0.2$, while fated sink cells were assigned $R = -0.2$. Cells not designated as source or sink were assigned $S = 0$ and $R = -1.0 \times 10^{-3}$. PBA next computes fate bias directly, producing a cell × fate bias matrix. We set the stochasticity, $D = 1.0$. Each cell in the training set was assigned a fate bias. To evaluate progenitor cells in the test set, we used a nearest neighbour graph built from the progenitor cells in the training set, subsequently mapping (using $k = 20$ neighbours) the fate biases generated for the cells in the training set to those in the test set. We also re-implemented the original PBA algorithm in PyTorch to accelerate downstream analyses (Fig. 2).

Dynamo. We adapted the author-published tutorial for scRNA-seq[9]. Briefly, this approach uses PCA-transformed embeddings as input and relies on the 'stochastic' model for computing expression dynamics. The function, 'dyn.tl.gene_wise_confidence' was used to identify and filter genes whose velocity contradicts the 'direction' of cell 'movement' based on user-provided initial and final states.

CellRank. We adapted the author-published tutorial for running CellRank using the RNA velocity kernel, which relies on precomputing RNA velocity values using the scVelo dynamical model[8,10]. Per the tutorial, the 'combined kernel' was created from a velocity kernel and connectivity kernel, weighted 0.80 and 0.20, respectively. $n = 10$ macrostates were used to compute the absorption probabilities towards terminal states or fates.

TIGON. We ran TIGON using the prescribed parameters in the published protocol and a modified version of the author-provided tutorial[17]. Briefly, we fit the model using the recommended parameters described in ref. 17. We halted the training procedure after 100 epochs. Subsequently, similar to scDiffEq and PRESCIENT, we used the model training checkpoint from epoch 100 to predict cell trajectories and fates from the 2,081 cells.

Linear regression. We used the scikit-learn implementation of linear regression, fitting the model to the PCA representation of the training set cells and their tabulated fate biases. To address the deterministic nature of linear regression in cases where progenitor cells exhibit tied fate proportions (for example, equal 0.5/0.5 splits between two fates), we applied small-magnitude Gaussian noise ($\sigma = 0.005 - 0.01$) to the ground truth fate bias matrix before training. This noise injection serves as a principled tie-breaking mechanism that reflects the inherent stochasticity of biological fate decisions and prevents systematic bias in fate assignment for ambiguous cases. We then predict the fate biases of the PCA representation of cells from the test set and compute accuracy scores using scikit-learn, where accuracy is determined by comparing the predicted dominant fate (highest probability) with the observed dominant fate for each evaluated cell.

Linear regression coefficient analysis. For each fate, we mapped the model coefficients to the original gene expression space by computing the matrix product of the principal component coefficients (shape: 1 fate × 50 principal components) and PCA component loadings (shape: 50 principal components × 2,447 genes). Gene importance scores were calculated as the absolute value of this mapping, representing each gene's contribution to fate prediction across all principal components. Genes were ranked by importance score for each fate, and the top predictive genes were examined for biological relevance to haematopoietic lineage specification.

Negative cross-entropy calculation (task 1). We evaluate the distributional performance of methods on fate prediction (task 1) using negative cross-entropy:

$$\mathrm{CE}(p, q) = -\sum_i p_i \log(q_i) \tag{16}$$

where P_i is the true probability of fate i and q_i is the predicted probability. This metric assesses how well the entire predicted probability distribution aligns with the ground truth fate distribution, providing a complementary evaluation to majority prediction accuracy.

Simulated cell annotation with approximate nearest neighbours. Simulated cells were labelled using approximate nearest neighbours (via Spotify's Annoy library) with 10 trees, 20 neighbours and Euclidean distance in 50-dimensional PCA space. The classifier was trained on labelled cells from the training dataset and applied to model-simulated cells.

Simulated cell fate annotation. Here, 2,000 trajectories per progenitor cell were simulated to the terminal time point. Cell labels of terminal states were generated with approximate nearest neighbours as described above, enabling tabulation of predicted fate probabilities for comparison with lineage-traced ground truth.

Benchmark task 2: time-point recovery

Following the previously described protocol, we fit both scDiffEq and PRESCIENT to the subset of data in days 2 and 6 for which lineage information was recorded. scDiffEq was trained for 100 epochs. PRESCIENT models were trained for 2,500 epochs. Evaluation of each model was performed by sampling 10,000 cells from day 2, with replacement, weighted by the empirically derived rate of cell proliferation and simulating one trajectory per cell. We then computed the Wasserstein distance via Sinkhorn divergence between the simulated cell populations at both day 4 and day 6 against the observed cell populations at these respective time points. PRESCIENT was evaluated at every 100 epochs. scDiffEq was evaluated at every epoch. For both models, we report the test error for the epoch with the minimum training error.

Benchmark task 3: in silico gene perturbations

Procedure. Following the previously described protocol, we introduced perturbations to 200 cells sampled from day 2 of the LARRY dataset[15]. For each gene perturbed, we set the z-scored expression to the indicated target value and retransformed the scaled expression matrix using the prefit PCA model (to the original data). These perturbed cells were used as input to the scDiffEq model for predicting future states. We then label the model-predicted latent states using an approximate nearest neighbour classifier fit to the original dataset, as described above. We used Welch's independent two-sided t-test to assess significance in comparing the fractions of neutrophil and monocyte fates across conditions (perturbed versus unperturbed). Each presented comparison was performed over ten seeds.

Comparison of in silico perturbations. We recomputed differential gene expression for each major haematopoietic lineage in the LARRY dataset (neutrophil, monocyte, basophil) by comparing progenitors naturally committed to each fate (as identified by lineage tracing) with uncommitted progenitors. Analyses were performed on log-normalized gene expression counts. For each pairwise comparison, genes were retained if expressed in at least 5% of cells in either group. Differential expression was assessed using the Wilcoxon rank-sum test ('scipy.stats.ranksums'), and P values were corrected for multiple testing using the Benjamini–Hochberg procedure to control the false discovery rate (FDR). The $\log_2$(fold-change) was computed as the $\log_2$ ratio of mean expression between groups, with a pseudocount of 1 added to each group's mean to stabilize estimates for lowly expressed genes. Genes with FDR-adjusted $P < 0.05$ were considered significantly differentially expressed. We aggregated the mean $\log_2$(fold-change) computed in the fate perturbation screens (described above) over five replicates. We took the intersection of these results with the significant genes from the above differential expression analysis. We then calculated Pearson correlation coefficients, with statistical significance assessed at $\alpha = 0.05$.

Negative control analysis with housekeeping genes. We intersected genes in the LARRY dataset with the Hounkpe housekeeping gene set from MSigDB ('HOUNKPE_HOUSEKEEPING_GENES'). Each gene in the gene set was independently subjected to the identical in silico overexpression protocol used for TFs ($z = 10$), and their effects on cell fate outcomes (neutrophil, monocyte, basophil) were quantified using the same statistical framework described above. We compared the distribution of effect sizes ($\log_2$ fold-changes in fate abundance) between housekeeping genes and known haematopoietic TFs using statistical tests (Mann–Whitney U-test or t-test, as appropriate).

Cosine divergence of drift fields

For a given cell with two predicted drift vector fields in 50-dimensional PCA space, we compared them by computing their cosine similarity.

Let $\mathbf{v}_i$ and $\mathbf{w}_i \in R^{50}$ denote the drift vectors for cell i from the reference and model fields, respectively. The cosine similarity is:

$$\cos(\theta_i) = \frac{\mathbf{v}_i \cdot \mathbf{w}_i}{|\mathbf{v}_i||\mathbf{w}_i|} \quad (17)$$

To then transformed cosine similarity, $\cos(\theta_i)$ to obtain the cosine divergence score, d as:

$$d_i = \frac{1 - \cos(\theta_i)}{2} \quad (18)$$

where d_i ranges from 0 (perfect alignment) to 1 (directionally inverted) with 0.5 corresponding to orthogonality. This metric emphasizes differences in vector field orientation although it remains scale-invariant to the magnitude of the velocity vectors.

Synthetic data generation

We generated datasets synthetically derived from the pancreatic endocrinogenesis dataset[43] using Gaussian mixture models. First, we fit a Gaussian mixture model with 20 components to the original 50-dimensional PCA space. We then sampled the Gaussian mixture model to generate synthetic cells that preserve the statistical properties of the biological data. Next, we assigned pseudotime values to the synthetic cells using k-nearest neighbours ($k = 5$) interpolation in PCA space with inverse distance weighting, implemented via the Annoy approximate nearest neighbour algorithm. Finally, we binned the pseudotime values into 2, 3, 5 or 10 discrete time points.

Computational complexity analyses

Using the protocol described in the section 'Synthetic data generation', we generated datasets of varying sizes (1,000, 10,000, 100,000 and 1,000,000 cells) and trajectory configurations (2time points spanning 1 day, 3 time points spanning 2 days, 5 time points spanning 4 days and 10 time points spanning 9 days). All remaining model hyperparameters were held constant: drift network (two hidden layers of 512 units), diffusion network (two hidden layers of 32 units), batch size (512 cells) and dt (0.1 d). Each configuration was trained over five independent initializations for 500 epochs to enable fair computational comparisons.

We measured peak GPU memory usage (reserved and allocated) during training using PyTorch CUDA memory tracking, training time (wall-clock time for 500 epochs), per-epoch timing (mean time per training iteration) and CPU memory (peak system RAM consumption during training). All training runs were conducted on NVIDIA T4 GPUs (~15 GB of memory) with identical hardware and software environments to ensure reproducible measurements.

Architecture-specific computational profiling was conducted using the LARRY dataset fate-prediction benchmark (task 1). We systematically varied drift and diffusion network architectures from small (two hidden layers of 32 units) to large (two hidden layers of 2,048 units), training each configuration for 2,500 epochs across five independent seeds. Batch sizes of 512 or 2,048 cells were tested to assess memory-performance tradeoffs. This analysis provides practical guidance for users to optimize model architecture within their computational constraints while maintaining biological performance.

Ablation studies

Ablation of diffusion. To ablate the scDiffEq's capacity to explicitly model cell-specific diffusion, we removed the diffusion network. Specifically, instead of parameterizing the diffusion term using a neural network that produces state-specific diffusion vectors, we set a fixed diffusion magnitude (calibrated to match the mean diffusion magnitude of the full model) that was constant across all cell states. Similar to PRESCIENT, this formulation results in a model operating as follows:

$$\Delta\mathbf{z}(t) = \mathbf{f}(\mathbf{z}(t))\Delta t + \Delta\mathbf{w}_t \quad (19)$$

where, as before, $\Delta\mathbf{w}_t$ represents an increment of Brownian motion over the time interval, Δt, with $w_{i,j}(t) \sim N(0, \Delta t)$. is a standard Wiener process (Brownian motion). All other components of the model, including the network architecture of the drift term, optimization procedure and other hyperparameters remained identical to the full model. This ablated model was trained and evaluated using the same procedures described for task 1 (fate prediction), with five random initializations.

Measuring the effect of temporal resolution on model performance. To evaluate how the number of available time points affects scDiffEq performance, we subset the LARRY dataset to neutrophil and monocyte lineages and used CytoTRACE to assign continuous developmental time to each cell, which was then binned into 6 discrete time points (0–5). We trained four scDiffEq model variants: v.0 using all time points [0,1,2,3,4,5], v.1 using [0,1,3,5], v.2 using [0,3,5] and v.3 using [0,5]. Each model was trained over 500 epochs and otherwise using the same hyperparameters as the main LARRY analysis. To evaluate each model, we sampled 10,000 trajectories from 137 initial progenitor cells (time bin 0) with replacement and computed the Sinkhorn divergence (with the same hyperparameters as previously described) between predicted and observed cell populations at each time point. This analysis allowed us to assess both interpolation accuracy (at unobserved time points) and reconstruction fidelity (at training time points) as a function of temporal resolution.

Pseudotime sensitivity analysis. To evaluate scDiffEq's robustness to pseudotime inaccuracies, we systematically degraded the quality of CytoTRACE-derived pseudotime values for the pancreatic endocrinogenesis dataset. For each degradation level (20%, 40%, 60%, 80% and 100%), we randomly selected the corresponding percentage of cells and shuffled their pseudotime values while maintaining the original values for the remaining cells. The degraded pseudotime values were then binned into discrete time points using the same protocol as the reference dataset. scDiffEq models were trained on each degraded dataset using fixed hyperparameters (drift network: two hidden layers of 512 units; diffusion network: two hidden layers of 32 units; batch size: 512 cells; dt: 0.1 d) for 2,500 epochs. The learned velocity fields were visualized using UMAP projections with drift represented as vector fields and diffusion magnitude shown as background colouring. This analysis enables qualitative assessment of how pseudotime accuracy affects the coherence and biological interpretability of learned developmental trajectories.

Statistical analyses and reproducibility

Software and hardware specifications. Neural networks were implemented in PyTorch. We used the Google/torchsde library for SDE integration. Models were trained on an NVIDIA Tesla T4 GPU with approximately 15 GB of memory, except for the large scDiffEq variant used in task 2 (time-point recovery), which was trained on an NVIDIA A100 GPU with approximately 40 GB of memory. The scDiffEq framework was implemented as a Python package compatible with the scverse ecosystem. scDiffEq is primarily built on PyTorch and PyTorch-Lightning.

Random seed management. For reproducibility, all experiments used 5–10 random seed initializations, indicated for specific instances. Random seeds controlled PyTorch random number generation, NumPy random states and Python's built-in random module. Deterministic CUDA operations were enabled where computationally feasible.

Statistical testing. Statistical significance was assessed using appropriate tests: Welch's two-sided t-test for perturbation analyses, Wilcoxon rank-sum test for differential expression and Mann–Whitney U-test for comparing effect size distributions. Multiple testing correction used the Benjamini–Hochberg procedure to control the FDR

(<0.05). Pearson correlation coefficients were calculated with significance assessed at $\alpha = 0.05$.

Box plots. For box plots used to display distributions of results in this paper, the mean result is represented by a dashed line. The boxes illustrate the interquartile range (IQR) (Q1 to Q3), while whiskers extend to the most extreme data point within 1.5 times the IQR from the box. Any points beyond this range are considered outliers and are shown as individual markers.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets used in this study were previously published and are publicly available. The LARRY dataset is available via GitHub at https://github.com/AllonKleinLab/paper-data/tree/master/Lineage_tracing_on_transcriptional_landscapes_links_state_to_fate_during_differentiation (commit: af842ce)[26]. The human haematopoiesis dataset, including the final list of filtered cells was provided by the authors of ref. 9 and is available under the GEO accession number GSE193517 as well as via direct download using the 'dynamo' Python package[9]. The pancreatic endocrinogenesis dataset is available via GitHub at https://github.com/theislab/scvelo_notebooks/raw/master/data/Pancreas/endocrinogenesis_day15.h5ad (commit: 2425e7f)[26,43]. All data generated in this study are available via Zenodo at https://doi.org/10.5281/zenodo.17238611 (ref. 53).

Code availability

An implementation of scDiffEq is available as a Python package via GitHub at https://github.com/scDiffEq/scDiffEq and https://pypi.org/project/scdiffeq/ (ref. 54). All analysis code, notebooks and scripts to reproduce the results presented in this paper are deposited in a dedicated reproducibility repository via GitHub at https://github.com/scDiffEq/scdiffeq-analyses (ref. 53). Comprehensive documentation is accessible at https://scdiffeq.com.

References

1. Qiu, X. et al. Reversed graph embedding resolves complex single-cell trajectories. *Nat. Methods* **14**, 979–982 (2017).
2. Street, K. et al. Slingshot: cell lineage and pseudotime inference for single-cell transcriptomics. *BMC Genomics* **19**, 477 (2018).
3. Chen, H. et al. Single-cell trajectories reconstruction, exploration and mapping of omics data with STREAM. *Nat. Commun.* **10**, 1903 (2019).
4. Saelens, W., Cannoodt, R., Todorov, H. & Saeys, Y. A comparison of single-cell trajectory inference methods. *Nat. Biotechnol.* **37**, 547–554 (2019).
5. Wang, S.-W., Herriges, M. J., Hurley, K., Kotton, D. N. & Klein, A. M. CoSpar identifies early cell fate biases from single-cell transcriptomic and lineage information. *Nat. Biotechnol.* **40**, 1066–1074 (2022).
6. Schiebinger, G. et al. Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming. *Cell* **176**, 1517 (2019).
7. La Manno, G. et al. RNA velocity of single cells. *Nature* **560**, 494–498 (2018).
8. Bergen, V., Lange, M., Peidli, S., Wolf, F. A. & Theis, F. J. Generalizing RNA velocity to transient cell states through dynamical modeling. *Nat. Biotechnol.* **38**, 1408–1414 (2020).
9. Qiu, X. et al. Mapping transcriptomic vector fields of single cells. *Cell* **185**, 690–711 (2022).
10. Lange, M. et al. CellRank for directed single-cell fate mapping. *Nat. Methods* **19**, 159–170 (2022).
11. Weiler, P., Lange, M., Klein, M., Pe'er, D. & Theis, F. CellRank 2: unified fate mapping in multiview single-cell data. *Nat. Methods* **21**, 1196–1205 (2024).
12. Gorin, G., Fang, M., Chari, T. & Pachter, L. RNA velocity unraveled. *PLoS Comput. Biol.* **18**, e1010492 (2022).
13. Weinreb, C., Wolock, S., Tusi, B. K., Socolovsky, M. & Klein, A. M. Fundamental limits on dynamic inference from single-cell snapshots. *Proc. Natl Acad. Sci. USA* **115**, E2467–E2476 (2018).
14. Hashimoto, T., Gifford, D. & Jaakkola, T. Learning population-level diffusions with generative RNNs. In *Proc. 33rd International Conference on Machine Learning* (eds Balcan, M. F. & Weinberger, K. Q.) Vol. 48, 2417–2426 (PMLR, 2016).
15. Yeo, G. H. T., Saksena, S. D. & Gifford, D. K. Generative modeling of single-cell time series with PRESCIENT enables prediction of cell trajectories with interventions. *Nat. Commun.* **12**, 3222 (2021).
16. Li, Q. scTour: a deep learning architecture for robust inference and accurate prediction of cellular dynamics. *Genome Biol.* **24**, 149 (2023).
17. Sha, Y., Qiu, Y., Zhou, P. & Nie, Q. Reconstructing growth and dynamic trajectories from single-cell transcriptomics data. *Nat. Mach. Intell.* **6**, 25–39 (2024).
18. Tong, A., Huang, J., Wolf, G., van Dijk, D. & Krishnaswamy, S. TrajectoryNet: a dynamic optimal transport network for modeling cellular dynamics. *Proc. Mach. Learn Res.* **119**, 9526–9536 (2020).
19. Huguet, G. et al. Manifold interpolating optimal-transport flows for trajectory inference. *Adv. Neural Inf. Process. Syst.* **35**, 29705–29718 (2022).
20. Elowitz, M. B., Levine, A. J., Siggia, E. D. & Swain, P. S. Stochastic gene expression in a single cell. *Science* **297**, 1183–1186 (2002).
21. Zechner, C., Nerli, E. & Norden, C. Stochasticity and determinism in cell fate decisions. *Development* **147**, dev181495 (2020).
22. Chen, R. T. Q., Rubanova, Y., Bettencourt, J. & Duvenaud, D. K. Neural ordinary differential equations. *Adv. Neural. Inf. Process. Syst.* **31**, 6572–6583 (2018).
23. Kidger, P., Foster, J., Li, X. & Lyons, T. J. Neural SDEs as infinite-dimensional GANs. In *Proc. 38th International Conference on Machine Learning* (eds Meila, M. & Zhang, T.) Vol. 139, 5453–5463 (PMLR, 2021).
24. Kidger, P. *On Neural Differential Equations*. DPhil thesis, University of Oxford (2021).
25. Jiang, Q. & Wan, L. A physics-informed neural SDE network for learning cellular dynamics from time-series scRNA-seq data. *Bioinformatics* **40**, ii120–ii127 (2024).
26. Weinreb, C., Rodriguez-Fraticelli, A., Camargo, F. D. & Klein, A. M. Lineage tracing on transcriptional landscapes links state to fate during differentiation. *Science* **367**, eaaw3381 (2020).
27. VanHorn, S. & Morris, S. A. Next-generation lineage tracing and fate mapping to interrogate development. *Dev. Cell* **56**, 7–21 (2021).
28. Alemany, A., Florescu, M., Baron, C. S., Peterson-Maduro, J. & van Oudenaarden, A. Whole-organism clone tracing using single-cell sequencing. *Nature* **556**, 108–112 (2018).
29. Biddy, B. A. et al. Single-cell mapping of lineage and identity in direct reprogramming. *Nature* **564**, 219–224 (2018).
30. Bowling, S. et al. An engineered CRISPR–Cas9 mouse line for simultaneous readout of lineage histories and gene expression profiles in single cells. *Cell* **181**, 1693–1694 (2020).
31. Frieda, K. L. et al. Synthetic recording and in situ readout of lineage information in single cells. *Nature* **541**, 107–111 (2017).
32. Raj, B. et al. Simultaneous single-cell profiling of lineages and cell types in the vertebrate brain. *Nat. Biotechnol.* **36**, 442–450 (2018).
33. Spanjaard, B. et al. Simultaneous lineage tracing and cell-type identification using CRISPR-Cas9-induced genetic scars. *Nat. Biotechnol.* **36**, 469–473 (2018).

34. Bunne, C. et al. Learning single-cell perturbation responses using neural optimal transport. *Nat. Methods* **20**, 1759–1768 (2023).
35. Giladi, A. et al. Single-cell characterization of haematopoietic progenitors and their trajectories in homeostasis and perturbed haematopoiesis. *Nat. Cell Biol.* **20**, 836–846 (2018).
36. Feinberg, M. W. et al. The Kruppel-like factor KLF4 is a critical regulator of monocyte differentiation. *EMBO J.* **26**, 4138–4148 (2007).
37. Yamanaka, R. et al. Impaired granulopoiesis, myelodysplasia, and early lethality in CCAAT/enhancer binding protein epsilon-deficient mice. *Proc. Natl Acad. Sci. USA* **94**, 13187–13192 (1997).
38. Serwas, N. K. et al. Mutant specific granule deficiency correlates with aberrant granule organization and substantial proteome alterations in neutrophils. *Front. Immunol.* **9**, 588 (2018).
39. Przybyla, L. & Gilbert, L. A. A new era in functional genomics screens. *Nat. Rev. Genet.* **23**, 89–103 (2022).
40. Hounkpe, B. W., Chenou, F., de Lima, F. & De Paula, E. V. HRT Atlas v1.0 database: redefining human and mouse housekeeping genes and candidate reference transcripts by mining massive RNA-seq datasets. *Nucleic Acids Res.* **49**, D947–D955 (2021).
41. Liberzon, A. et al. The Molecular Signatures Database (MSigDB) hallmark gene set collection. *Cell Syst.* **1**, 417–425 (2015).
42. Qiu, Q. et al. Massively parallel and time-resolved RNA sequencing in single cells with scNT-seq. *Nat. Methods* **17**, 991–1001 (2020).
43. Bastidas-Ponce, A. et al. Comprehensive single cell mRNA profiling reveals a detailed roadmap for pancreatic endocrinogenesis. *Development* **146**, dev173849 (2019).
44. Gulati, G. S. et al. Single-cell transcriptional diversity is a hallmark of developmental potential. *Science* **367**, 405–411 (2020).
45. Liggett, L. A. & Sankaran, V. G. Unraveling hematopoiesis through the lens of genomics. *Cell* **182**, 1384–1400 (2020).
46. Dahl, R., Iyer, S. R., Owens, K. S., Cuylear, D. D. & Simon, M. C. The transcriptional repressor GFI-1 antagonizes PU.1 activity through protein-protein interaction. *J. Biol. Chem.* **282**, 6473–6483 (2007).
47. Orkin, S. H. & Zon, L. I. Hematopoiesis: an evolving paradigm for stem cell biology. *Cell* **132**, 631–644 (2008).
48. Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M. & Le, M. Flow matching for generative modeling. *Int. Conf. Learn. Represent.* (ICLR, 2023).
49. Tong, A. et al. Simulation-free Schrödinger bridges via score and flow matching. *Proc. 27th Int. Conf. Artif. Intell. Stat. (AISTATS)* **238**, 1279–1287 (2024).
50. Virshup, I. et al. The scverse project provides a computational ecosystem for single-cell omics data analysis. *Nat. Biotechnol.* **41**, 604–606 (2023).
51. Feydy, J. et al. Interpolating between optimal transport and MMD using Sinkhorn divergences. In *Proc. Twenty-Second International Conference on Artificial Intelligence and Statistics* (eds Chaudhuri, K. & Sugiyama, M.) Vol. 89, 2681–2690 (PMLR, 2019).
52. Satija, R., Farrell, J. A., Gennert, D., Schier, A. F. & Regev, A. Spatial reconstruction of single-cell gene expression data. *Nat. Biotechnol.* **33**, 495–502 (2015).
53. Vinyard, M. scDiffEq/scdiffeq-analyses: v.0.1.0rc0. *Zenodo* https://doi.org/10.5281/zenodo.17238611 (2025).
54. Vinyard, M. scDiffEq/scDiffEq: v0.1.10rc0. Zenodo https://doi.org/10.5281/zenodo.17238594 (2025).

Acknowledgements

We graciously thank C. Weinreb for their assistance in using the LARRY dataset. We thank S. Saksena, G. H. Ting Yeo and D. K. Gifford for their assistance in creating a benchmark comparison to PRESCIENT[15]. We thank X. Qiu for their assistance in comparing scDiffEq with Dynamo and using the human haematopoiesis scNT-seq dataset. We thank the entire Pinello and Getz Laboratories for thoughtful feedback and discussion throughout the preparation of this paper. We thank S. Anand for pivotal discussions that guided us towards the useful implementation of neural differential equations. We thank A. Ravi for useful conceptual design discussions. We thank D. Jia, J. Zhou and H. Levine for useful conversations and assistance in simulating synthetic datasets for proof-of-concept experiments. We thank M. Miller for editorial assistance. While working on this paper, M.E.V. was supported by the National Institutes of Health (NIH) under the Ruth L. Kirschstein National Research Service Award (grant no. 1F31CA257625) from the National Cancer Institute (NCI). L.P. was supported by the National Institutes of Health (NIH) Genomic Innovator Award (grant no. 1R35HG010717-01) and the Rappaport MGH Research Scholar Award 2024-2029. G.G. was partially funded by the Paul C. Zamecnik Chair in Oncology at the Mass General Cancer Center.

Author contributions

All authors contributed to the conceptualization of the methodology, experiments and analyses. M.E.V. performed the M.E.V., A.W.R. and R.L. performed the fate-prediction benchmarking experiments. M.E.V. performed the time-point interpolation experiments. M.E.V. performed the gene perturbation experiments. M.E.V. and R.L. performed the investigation of model attributes and gene-level analyses. M.E.V. wrote the software package. All authors wrote the paper. A.M.K. provided guidance and edited the paper. G.G. and L.P. provided supervision and guidance in designing the experimental strategy and funded the research.

Competing interests

G.G. receives research funds from IBM, Pharmacyclics/Abbvie, Bayer, Genentech, Calico, Ultima Genomics, Inocras, Google, Kite and Novartis and is an inventor on patent applications filed by the Broad Institute related to MSMuTect, MSMutSig, POLYSOLVER, SignatureAnalyzer-GPU, MSEye, MinimuMM-seq and DLBclass. He is a founder and consultant and holds privately held equity in Scorpion Therapeutics; he is also a founder of, and holds privately held equity in, Predicta Biosciences; and holds privately held equity in Antares Therapeutics. M.E.V. is a founder of, and holds privately held equity in, Quintessence Laboratories, Inc. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s42256-025-01150-3.

Correspondence and requests for materials should be addressed to Gad Getz or Luca Pinello.

Peer review information *Nature Machine Intelligence* thanks Xiuwei Zhang and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2025

nature portfolio

Corresponding author(s): Luca Pinello (primary), Gad Getz

Last updated by author(s): 10/03/2025

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Python in conjunction with open-source libraries including wget, requests, scanpy, cellrank, and scvelo, and dynamo were used to obtain the publicly-available data.
Data analysis	An implementation of scDiffEq is available as a Python package at https://github.com/scDiffEq/scDiffEq and https://pypi.org/project/scdiffeq/. Notebooks and scripts to reproduce the results presented in this manuscript have been deposited to a dedicated repository at: https://github.com/scDiffEq/scdiffeq-analyses. Comprehensive documentation for the scDiffEq package is accessible at https://scdiffeq.com. All open-source dependencies of this software are detailed within the repository. There were no private/commercial softwares used for analysis.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All datasets used in this study were previously published and publicly available. The LARRY dataset was downloaded from: https://github.com/AllonKleinLab/paper-data/tree/master/Lineage_tracing_on_transcriptional_landscapes_links_state_to_fate_during_differentiation (commit: af842ce). The human hematopoiesis dataset, including the final list of filtered cells was provided by the authors of Qiu, et al., 2022 and is available under the GEO accession: GSE193517 as well as direct download using the `dynamo` python package. The pancreatic endocrinogenesis dataset was downloaded from: https://github.com/theislab/scvelo_notebooks/raw/master/data/Pancreas/endocrinogenesis_day15.h5ad (commit: 2425e7f).

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Not applicable to this study.
Reporting on race, ethnicity, or other socially relevant groupings	Not applicable to this study.
Population characteristics	Not applicable to this study.
Recruitment	Not applicable to this study.
Ethics oversight	Not applicable to this study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes (e.g., number of models tested) were determined based on the precedent of works we compare to.
Data exclusions	Not applicable to this study.
Replication	Model training was replicated with N=5. Replication attempts were successful.
Randomization	Not relevant to this study; using datasets that were collected in other previously published works.
Blinding	Blinding is not relevant to this study because we focused on training deep learning models with data that was collected and labeled in previously published works.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Not applicable to this study.
Novel plant genotypes	Not applicable to this study.
Authentication	Not applicable to this study.